

Fuzzy Inference System

KK
Prodi S1 RKA



Formal Languages

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

Figure 8.1 Formal languages and their ontological and epistemological commitments.

Fuzzy Logic

- Fuzzy Logic (FL) is a computational paradigm that is based on how humans think.
- The human brain can reason with uncertainties, vagueness, and judgments. Computers can only manipulate precise valuations. Fuzzy logic is an attempt to combine the two techniques (reasoning uncertainties and manipulate precise valuations).
- Fuzzy Logic looks at the world in imprecise terms, in much the same way that our brain takes in information (e.g. temperature is hot, speed is slow), then responds with precise actions.

Fuzzy Logic (2)

- FL is in fact, a precise problem-solving methodology.
- It is able to simultaneously handle numerical data and linguistic knowledge.
- Fuzzy logic differs from classical logic in that statements are no longer black or white, true or false, on or off.
- In fuzzy logic, a statement can assume any real value between 0 and 1, representing the degree to which an element belongs to a given set.

Fuzzy Logic Application (1)

- Fuzzy Logic enables low cost microcontrollers to perform functions traditionally performed by more powerful expensive machines enabling lower cost products to execute advanced features.

Intel Corporation's Embedded Microcomputer Division Fuzzy Logic Operation

MCS® 96/296 Microcontrollers
Designed to Meet Your Needs



The advertisement features a central image of an Intel microcontroller chip. To its right, there are two columns of links for the MCS 96 and MCS 296 families. The MCS 96 column includes links for Overview, HSIO Family, EPA Family, Motor Control Family, and CAN Product Family (Express). The MCS 296 column includes links for Overview, Background, and Documentation. The Intel logo is at the bottom right of the central image area.

MCS 96	MCS 296
- Overview - HSIO Family - EPA Family - Motor Control Family - CAN Product Family (Express)	- Overview - Background - Documentation

Motorola 68HC12 MCU

<http://www.intel.com/design/mcs96/designex/2351.htm>

men

Fuzzy Logic Application (2)

- In the city of Sendai in Japan, a **16-station subway system** is controlled by a fuzzy computer (Seiji Yasunobu and Soji Miyamoto of Hitachi) – the ride is so smooth, riders do not need to hold straps.
- **Nissan** – fuzzy automatic transmission, fuzzy anti-skid braking system
- **CSK, Hitachi** – Hand-writing Recognition
- **Tokyo's stock market** has had at least one stock-trading portfolio based on Fuzzy Logic that outperformed the Nikkei exchange average
- **NASA** has studied fuzzy control for automated space docking: simulations show that a fuzzy control system can greatly reduce fuel consumption
- **Canon** developed an auto-focusing camera that uses a charge-coupled device (CCD) to measure the clarity of the image in six regions of its field of view and use the information provided to determine if the image is in focus. It also tracks the rate of change of lens movement during focusing, and controls its speed to prevent overshoot.

Fuzzy Logic Application (3)

- For washing machines, Fuzzy Logic control is almost becoming a standard feature
- fuzzy controllers to load-weight, fabric-mix, and dirt sensors and automatically set the wash cycle for the best use of power, water, and detergent.

GE WPRB9110WH Top Load Washer

Haier ESL-T21 Top Load Washer

LG WD14121 Front Load Washer

Miele WT945 Front Load All-in-One Washer / Dryer

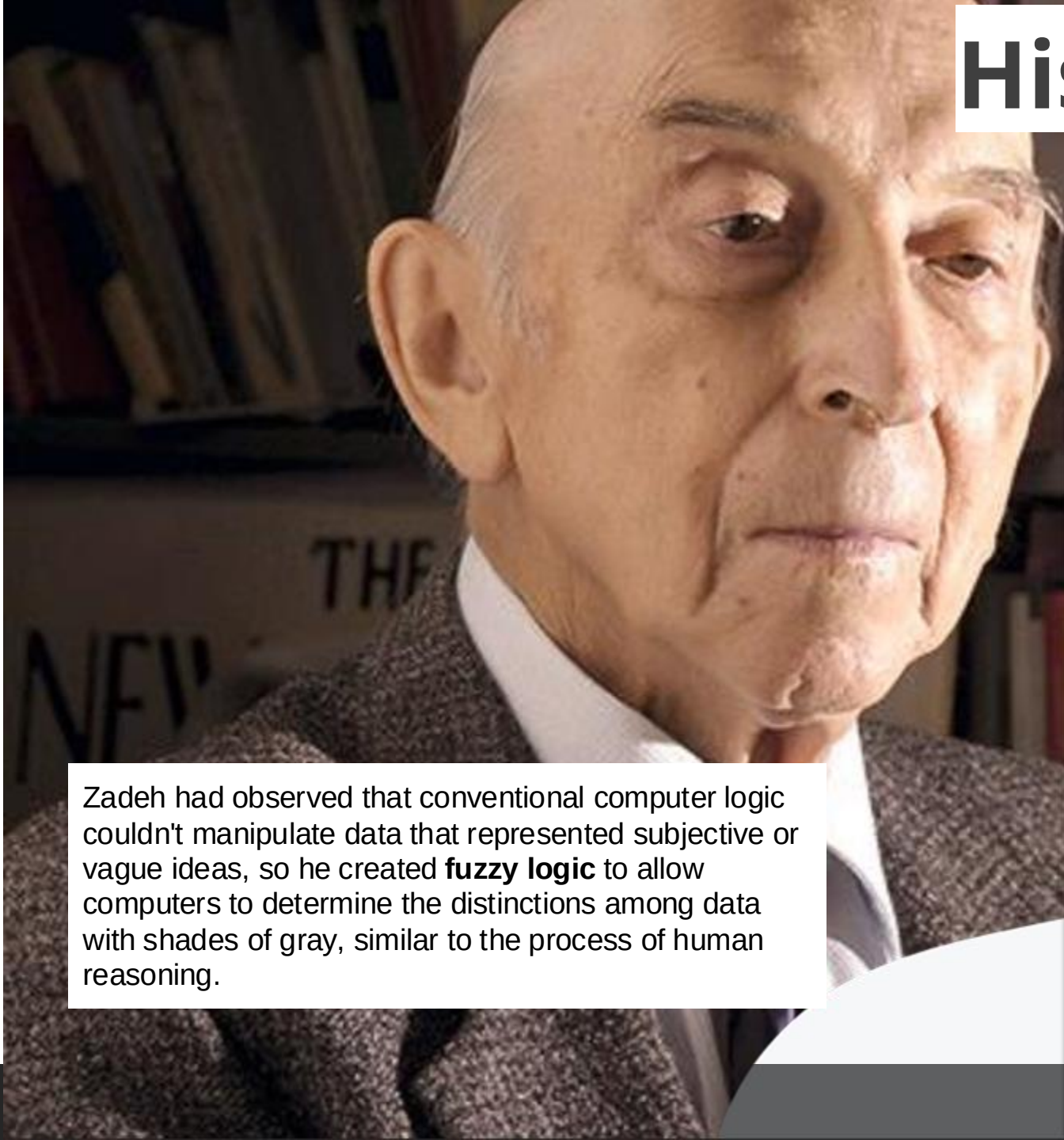
AEG LL1610 Front Load Washer

Zanussi ZWF1430W Front Load Washer

Others: Samsung, Toshiba, National, Matsushita, etc.



History of Fuzzy Logic

A close-up portrait of Lotfi A. Zadeh, an elderly man with white hair, wearing a dark suit and tie. He is looking slightly to the right. The background is a bookshelf with books.

Zadeh had observed that conventional computer logic couldn't manipulate data that represented subjective or vague ideas, so he created **fuzzy logic** to allow computers to determine the distinctions among data with shades of gray, similar to the process of human reasoning.

Precision and Meaning

"As complexity rises, precise statements lose **meaning** and meaningful statements lose **precision**."

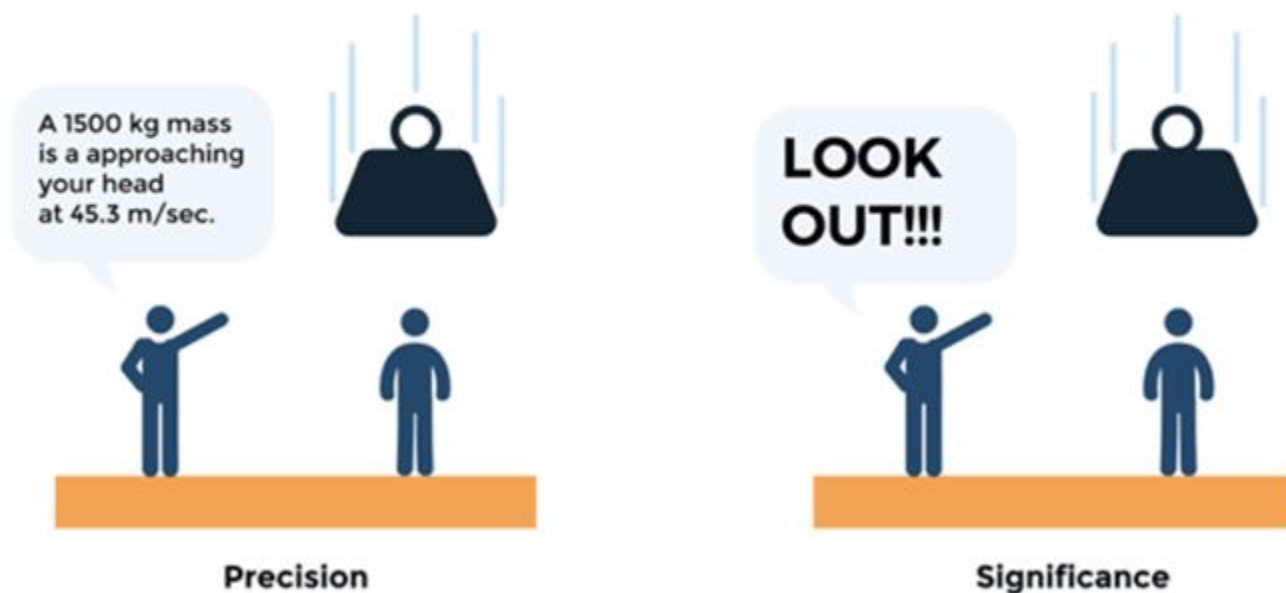
Lotfi A. Zadeh, 1965

<http://www.cs.berkeley.edu/~zadeh/>

Precision and Meaning

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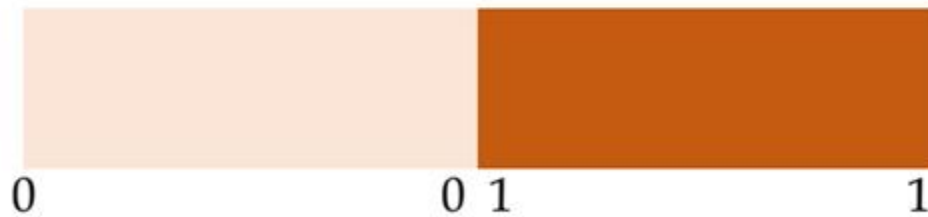


Fuzzy Sets

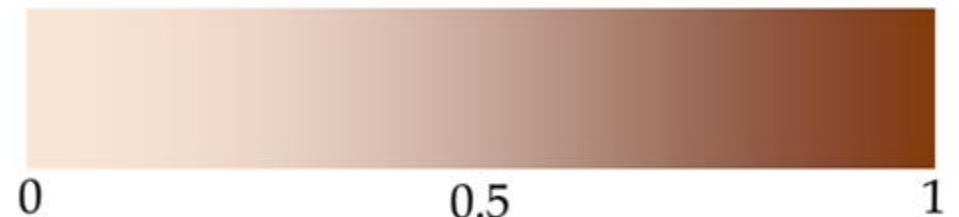
"The theory of fuzzy sets is **a theory of graded concepts**, a theory in which everything is a matter of degree."

Lotfi A. Zadeh, 1965

Unlike two-valued Boolean logic, fuzzy logic is based on **degrees of membership**. It deals with **degrees of truth**.

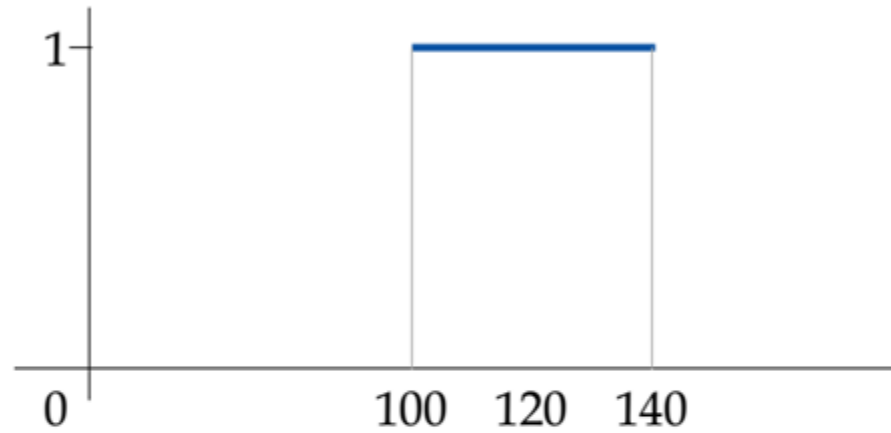


(a) Boolean Logic.



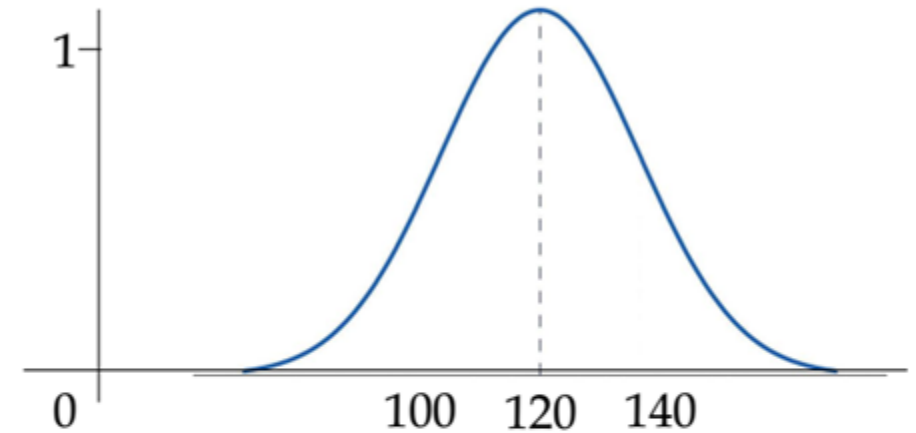
(b) Multi-valued Logic.

Fuzzy Sets



"normal blood pressure"
as (ordinary) set

VS.



"normal blood pressure"
as fuzzy set

replace $\{0, 1\}$ by $[0, 1]$

i.e., degrees of truth instead of false and true

Crisp Logic

- **Crisp logic** is concerned with absolutes-true or false, there is no in-between.

- Example:

Rule:

If the temperature is higher than 80°F, it is hot; otherwise, it is not hot.

Cases:

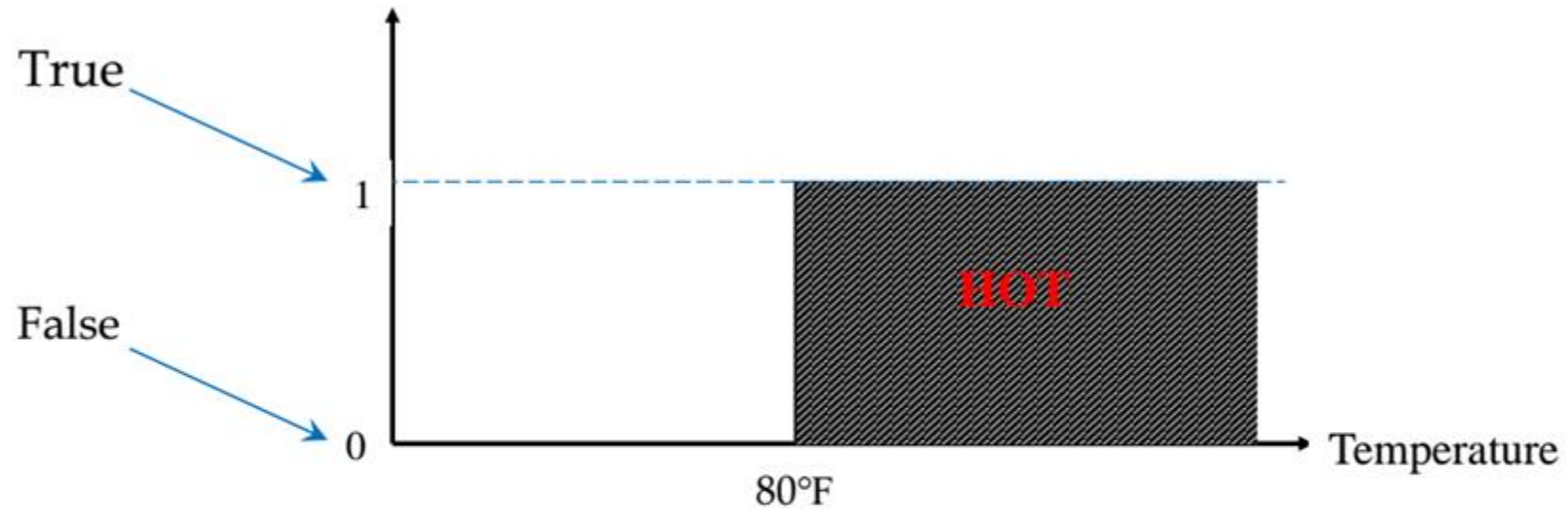
Temperature = 100°F **Hot**

Temperature = 80.1°F **Hot**

Temperature = 79.9°F **Not Hot**

Temperature = 50°F **Not Hot**

Crisp Set



If temperature $\geq 80^{\circ}\text{F}$, it is hot (1 or true);

If temperature $< 80^{\circ}\text{F}$, it is not hot (0 or false).

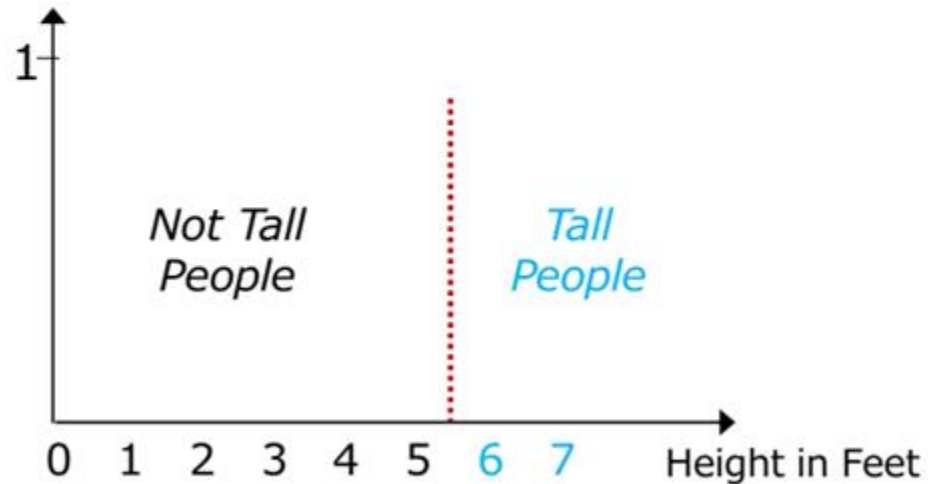
Crisp vs. Fuzzy Set

Set of tall men

covers all men, but their **degrees of membership** depend on their height.

Crisp Logic

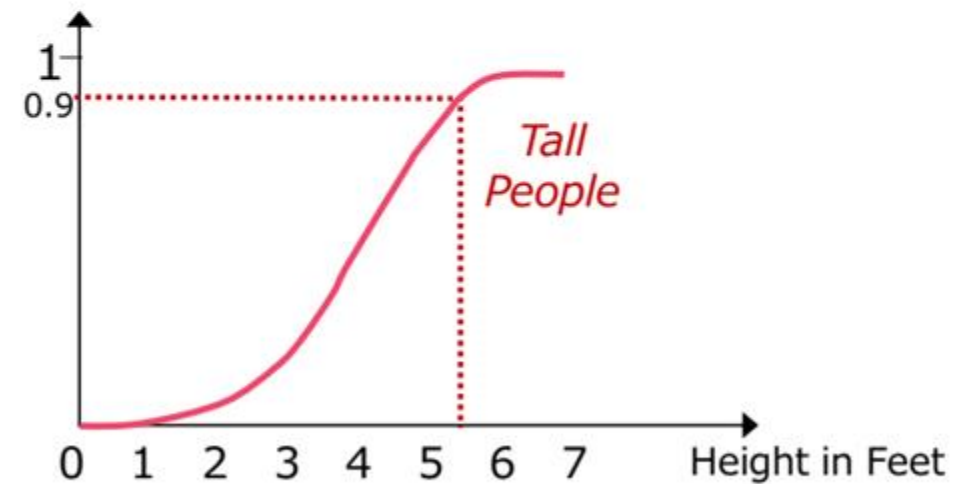
Crisp Set



Crisp membership

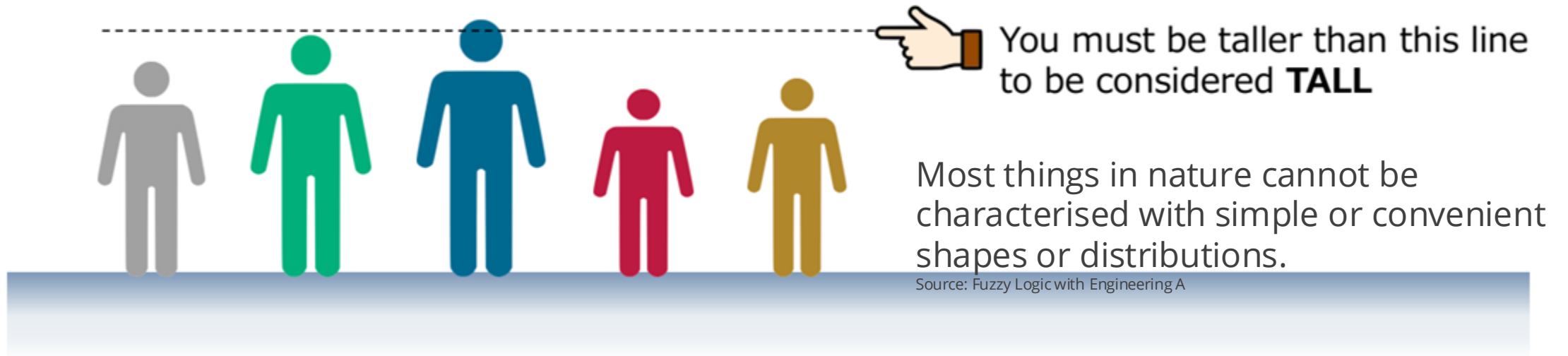
Fuzzy Logic

Fuzzy Set



Fuzzy membership

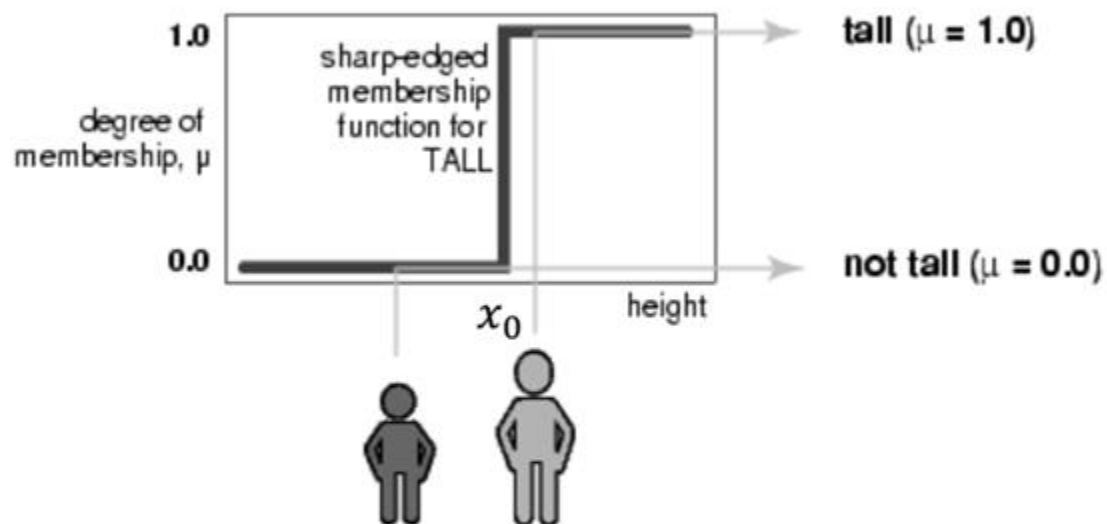
Membership Function



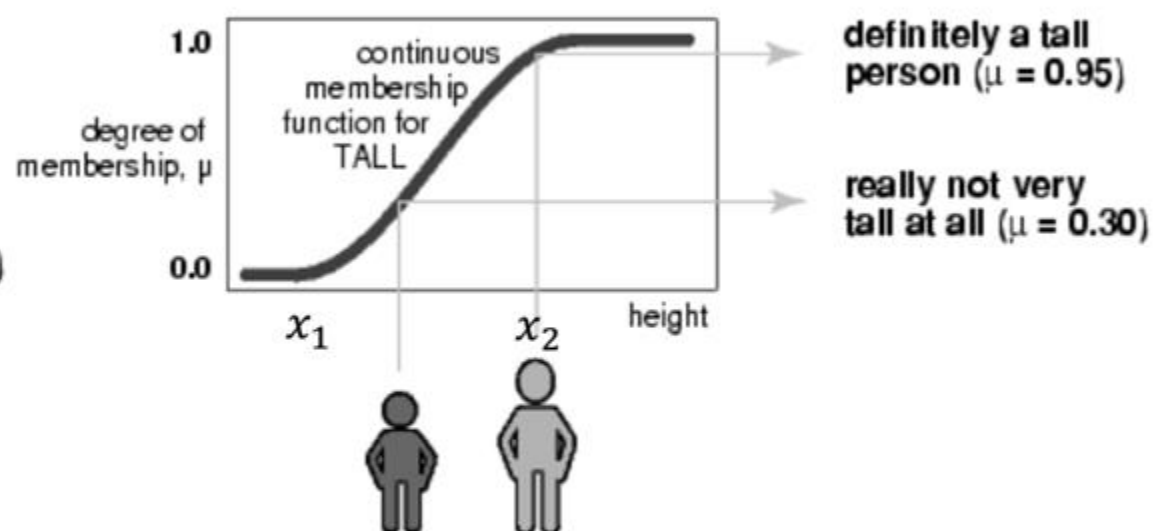
A **membership function** (MF) is a curve that defines how each point in the input space is mapped to a membership value (or degree of membership) between 0 and 1.

Membership functions characterize the fuzziness in a fuzzy set

Membership Function



$$\mu(x) = \begin{cases} 0 & \text{if } x < x_0, \\ 1 & \text{if } x \geq x_0 \end{cases}$$



$$\mu(x) = \begin{cases} 0 & \text{if } x \leq x_1, \\ 0 < \mu(x) < 1 & \text{if } x_1 < x < x_2, \\ 1 & \text{if } x \geq x_2 \end{cases}$$

Fuzzy Inference System

Fuzzy Inference System

- A fuzzy inference system (FIS) is also known as fuzzy-rule-based system, fuzzy expert system, fuzzy logic system, fuzzy model, fuzzy associative memory (FAM) and fuzzy logic controller and fuzzy system.
- An FIS is a computing framework based on the concepts of fuzzy set theory, fuzzy (If-Then) rules and fuzzy reasoning
- An FIS consists of 4 components: fuzzifier, Knowledge base (rule base or database), fuzzy inference engine and defuzzifier

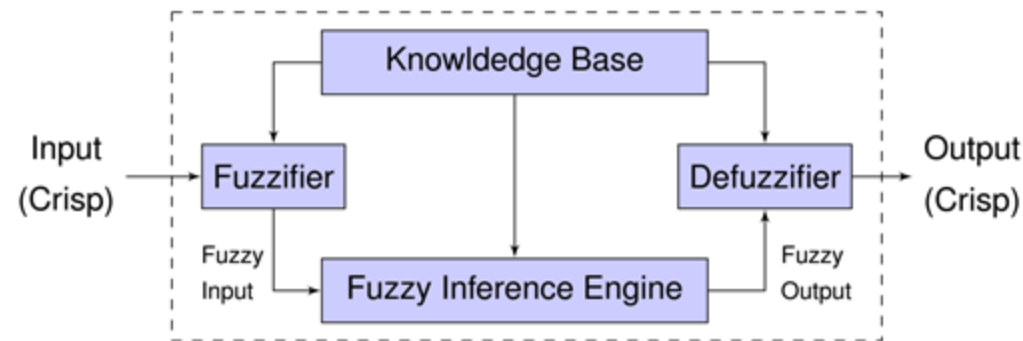


Figure 10: A diagram of fuzzy inference system.

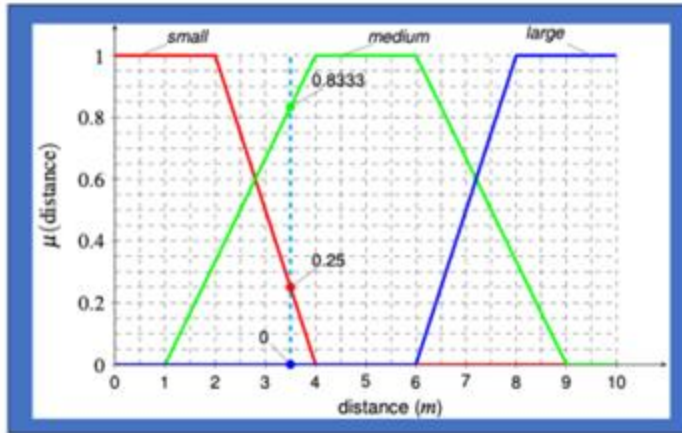
Fuzzy Inference System (2)

- **Fuzzifiers:** It maps the crisp (real-valued) input into a fuzzy set defined in the universe of discourse (the domain of the fuzzy set) X characterised by membership functions. This process is called fuzzification. Note: The input can also be a fuzzy set
- **Knowledge Base:** It is a database consisting of linguistic rules in If-Then format.
- **Fuzzy Inference Engine:** Using the If-Then rules in Knowledge base, it performs reasoning by producing a fuzzy output according to the fuzzy input given by the fuzzifier.
- **Defuzzifiers:** It converts the fuzzy output given by the fuzzy inference engine to produce a crisp (real-valued) output. This process is called defuzzification.

Example

Driving problem: I am driving and want to keep a safety distance between cars.
When the distance from the front car is x , what speed should I keep?

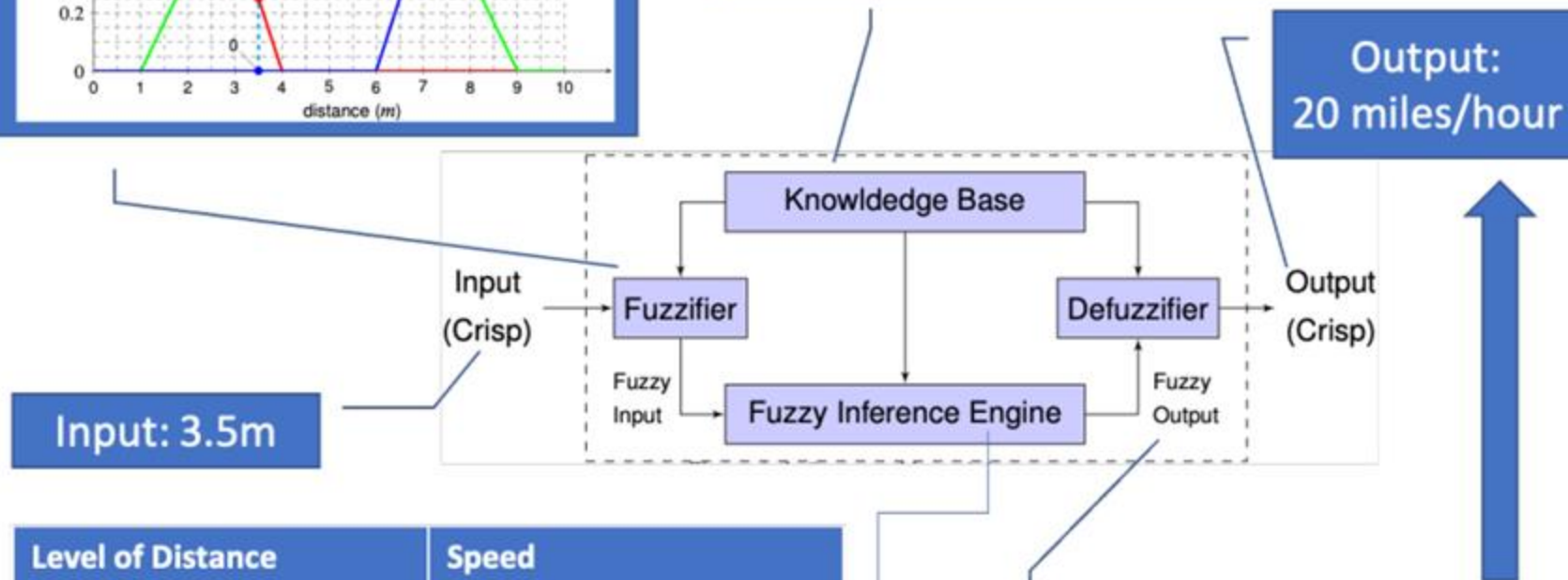




Rule 1: If distance is *small* Then speed is *low*

Rule 2: If distance is *medium* Then speed is *steady*

Rule 3: If distance is *large* Then speed is *high*



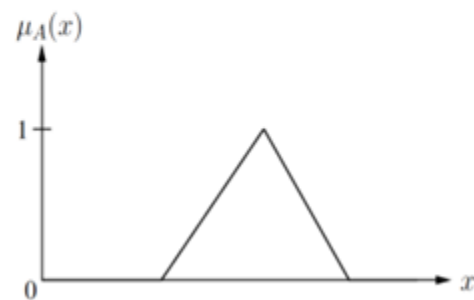
Level of Distance	Speed
0.25 of Small	Not very Low
0.8333 of Medium	More Steady
0 of Large	Definitely not High

Speed should **not** be **very Low**, **more** towards **Steady** but **definitely not High**

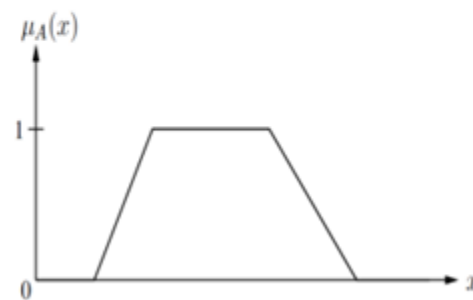
Fuzzification (Fuzzifier)

- The input is turned to fuzzy sets through membership functions.
- A fuzzy set is represented by a membership function (associated with a label called fuzzy term or linguistic term).
- There are various methods to assign membership values or the membership functions to fuzzy variables.

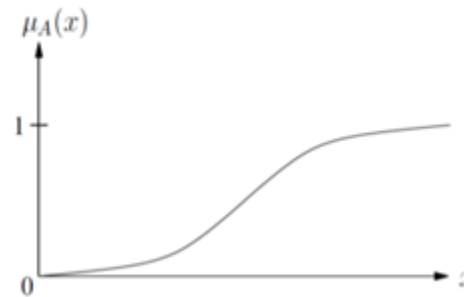
Membership functions for fuzzy sets can be of any shape or type as determined by experts in the domain over which the sets are defined



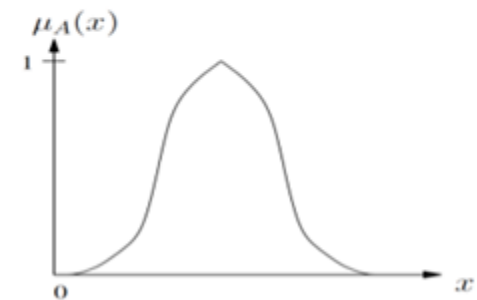
Triangular Function



Trapezoidal Function



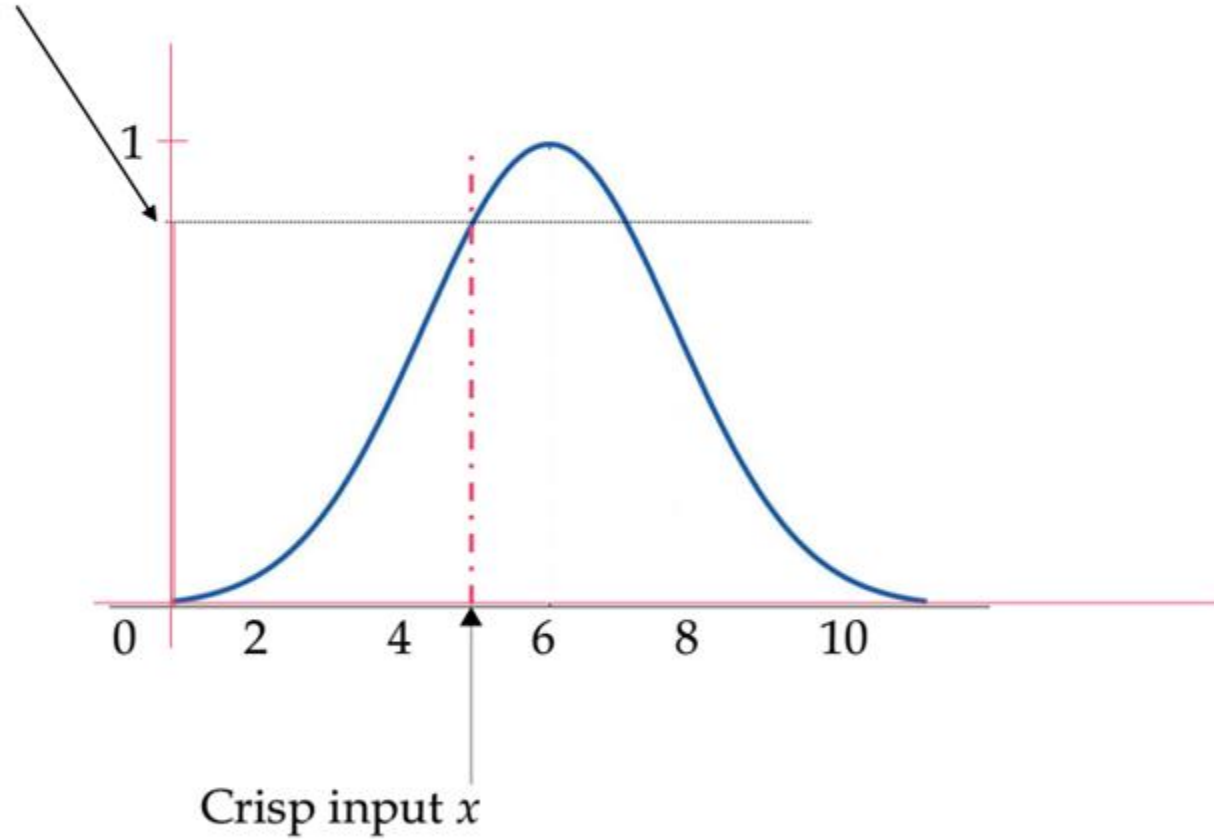
Logistic Function



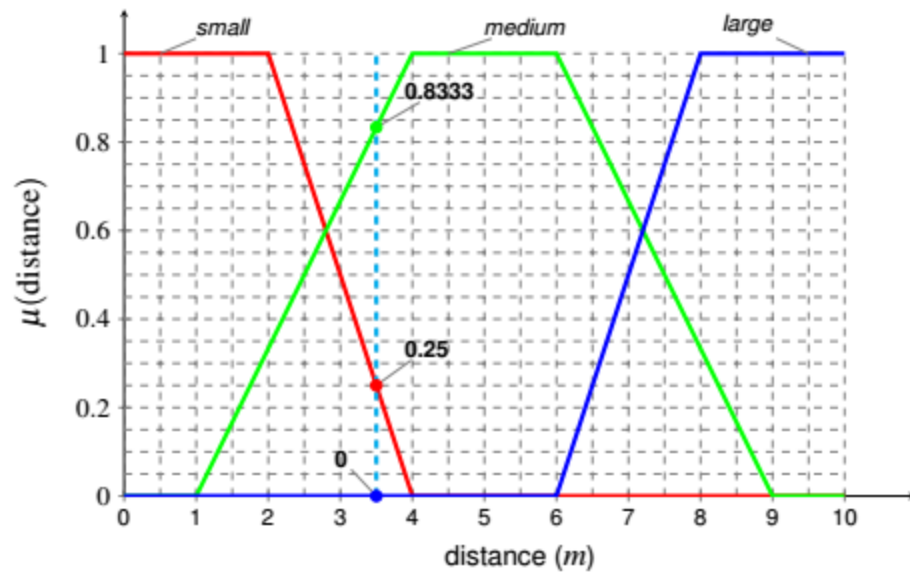
Gaussian Function

Fuzzification

Membership grade of crisp input x in the fuzzy set



Fuzzification



$$\mu = [\mu_s, \mu_M, \mu_L]$$

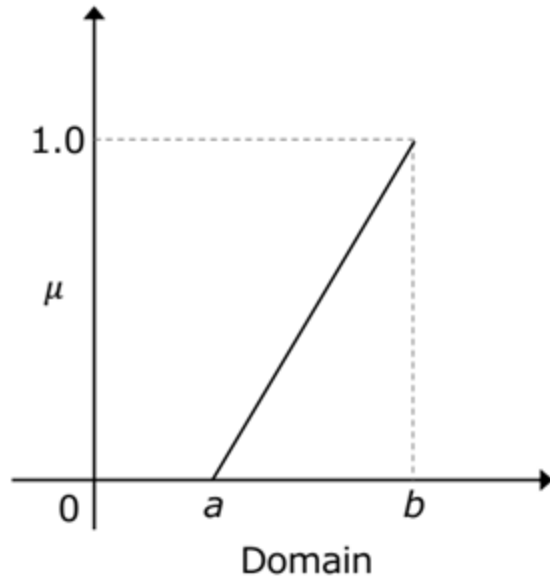
$$\mu[3.5] = [0.25, 0.833, 0]$$

$$\mu_s[3.5] = 0.25$$

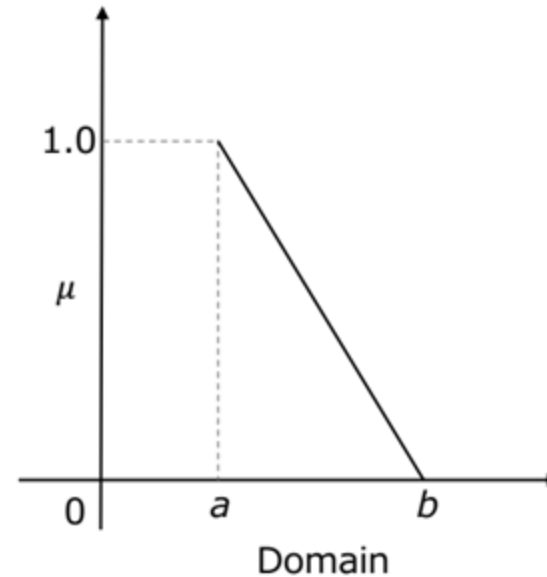
$$\mu_M[3.5] = 0.833$$

Fuzzification – How to compute membership value?

Membership Function: Linear Function

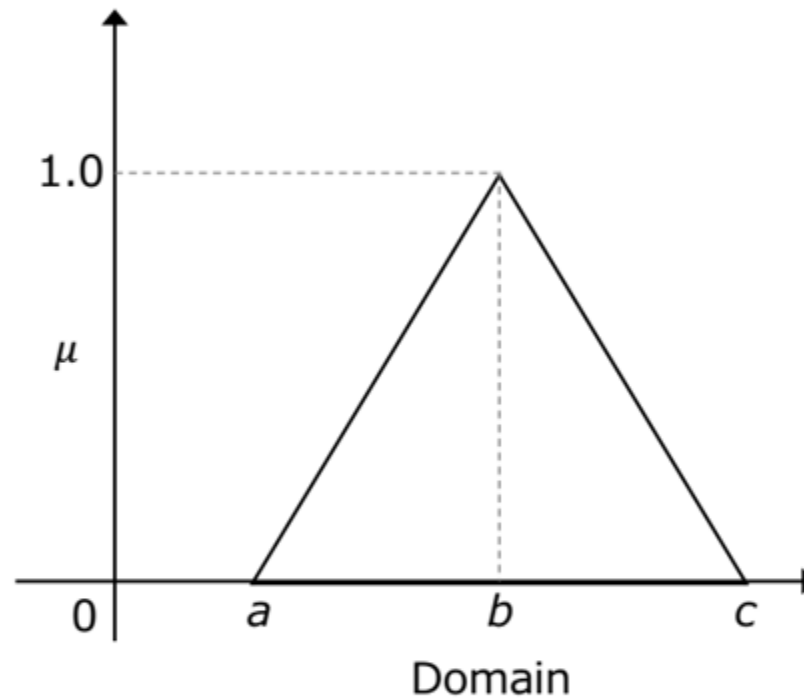


$$\mu[x] = \begin{cases} 0; & x \leq a \\ (x - a) / (b - a); & a < x \leq b \\ 1; & x > b \end{cases}$$



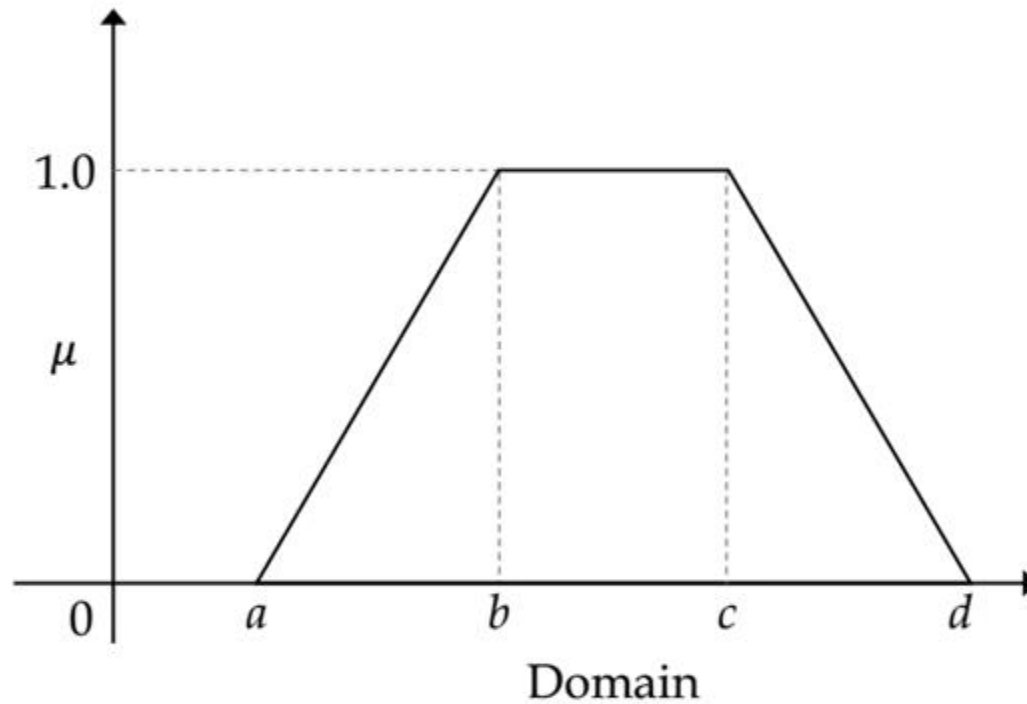
$$\mu[x] = \begin{cases} 1; & x \leq a \\ (b - x) / (b - a); & a \leq x < b \\ 0; & x \geq b \end{cases}$$

Membership Function: Triangular Function



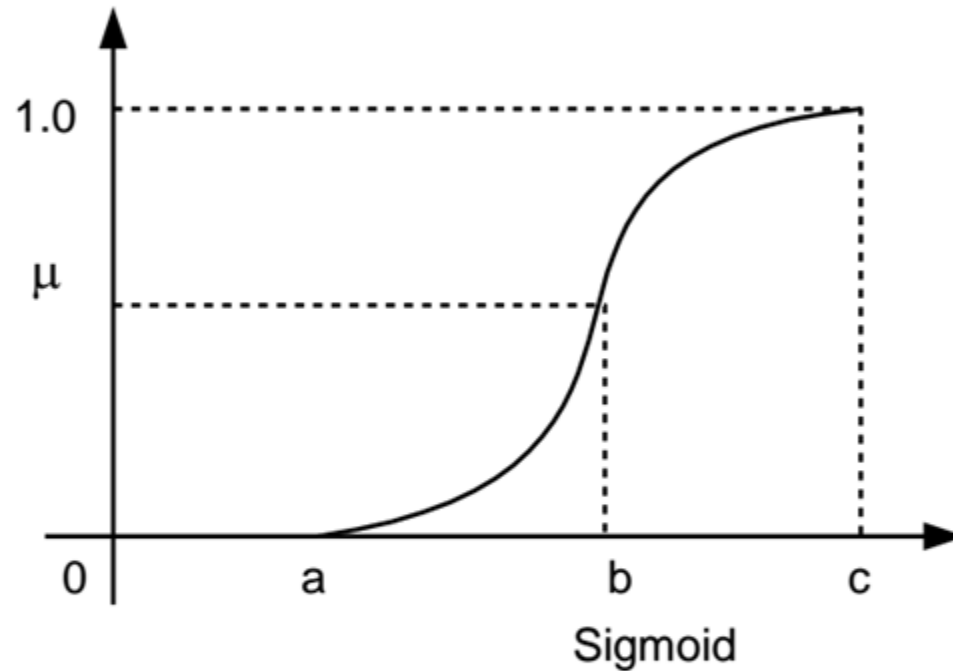
$$\begin{aligned}\mu[x] &= 0; \quad x \leq a \text{ or } x \geq c \\ &= (x - a) / (b - a); \quad a < x \leq b \\ &= (c - x) / (c - b); \quad b < x < c\end{aligned}$$

Membership Function: Trapezoidal Function



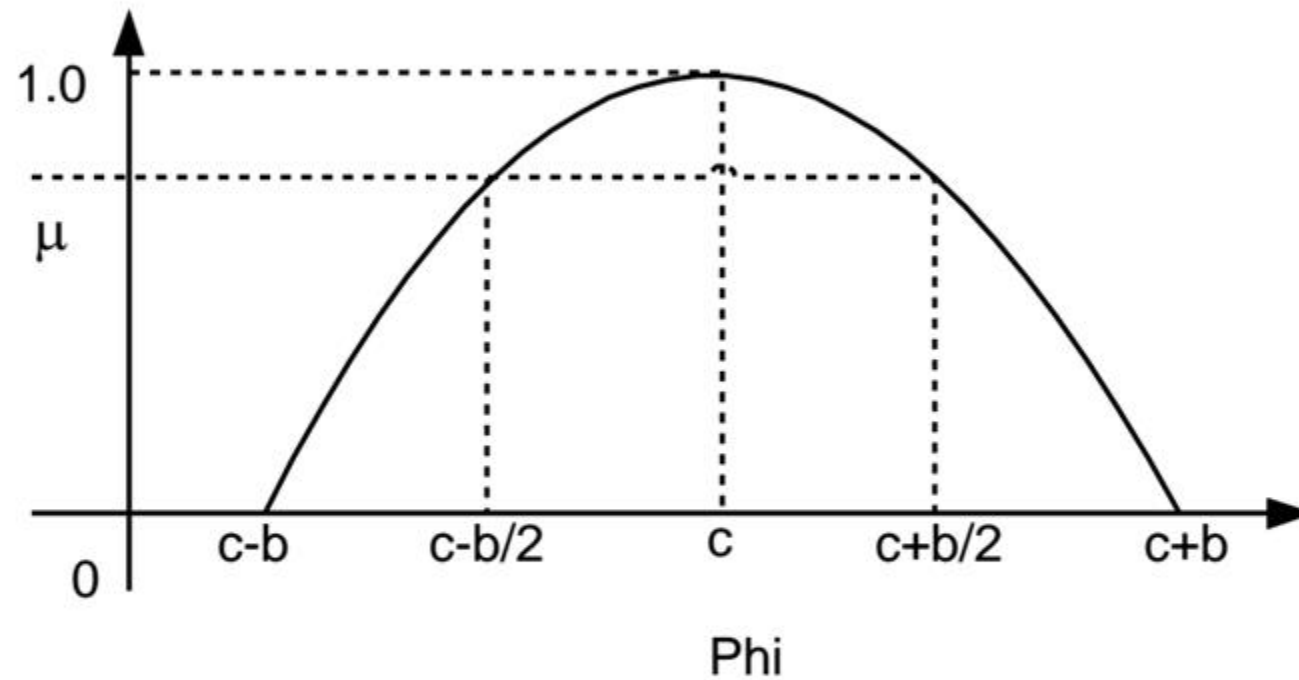
$$\begin{aligned}\mu[x] = & 0; x \leq a \text{ atau } x \geq d \\ & (x - a) / (b - a); a < x \leq b \\ & 1; b < x \leq c \\ & (d - x) / (d - c); c < x < d\end{aligned}$$

Membership Function: Sigmoid Function



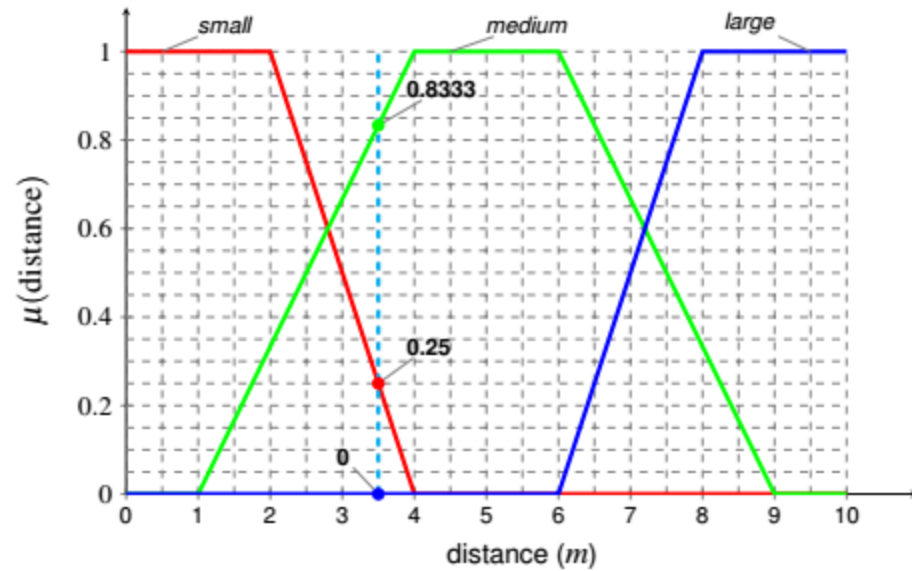
$$\mu[x; a, b, c]_{\text{sigmoid}} = \begin{cases} 0; & x \leq a \\ 2 \left(\frac{(x-a)}{(c-a)} \right)^2; & a < x \leq b \\ 1 - 2 \left(\frac{(c-x)}{(c-a)} \right)^2; & b < x < c \\ 1; & x \geq c \end{cases}$$

Membership Function: Phi Function



$$\begin{aligned}\mu[x; a, b, c]_{\text{phi}} &= \mu[x; c - b, c - b/2, c]_{\text{sigmoid}}; \quad x \leq c \\ &\mu[x; c, c + b/2, c + b]_{\text{sigmoid}}; \quad x > c\end{aligned}$$

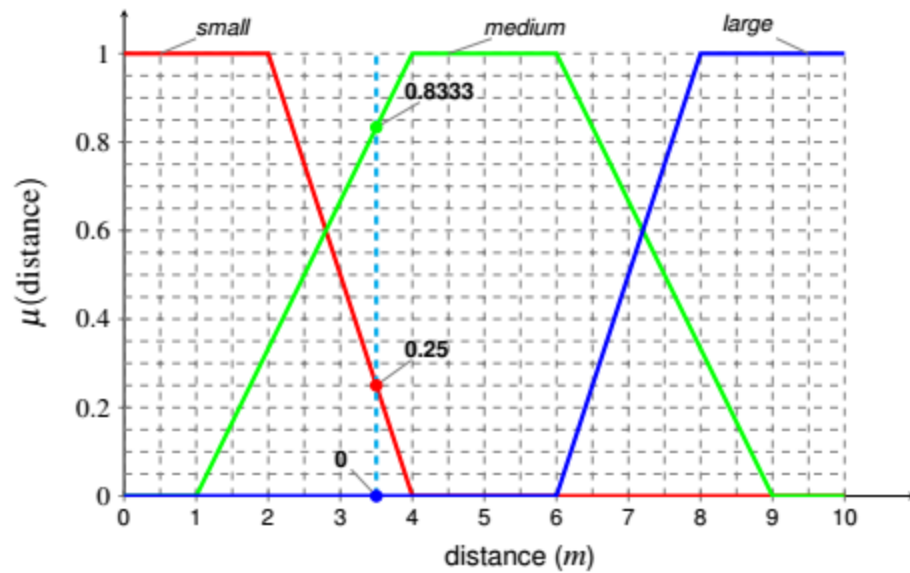
Practice



Compute a set of membership function if the distance is 2.5 m

Recall last week

- The output of fuzzification process is fuzzy set
- Example:



$$\mu = [\mu_s, \mu_m, \mu_l]$$

$$\mu[3.5] = [0.25, 0.833, 0]$$

$$\mu_s[3.5] = 0.25$$

$$\mu_m[3.5] = 0.833$$

fuzzy set

Knowledge Base

- The knowledge base is the rule base representing the expertise knowledge dealing with a specific problem.
- It is represented by linguistic rule: **IF** premise (antecedent) **THEN** conclusion (consequent).
- The knowledge base can have more than one rule

General rule format:

Rule i : IF x_1 is $\tilde{A}_{i1}$ and/or x_2 is $\tilde{A}_{i2}$ and/or $\dots$ THEN y is $\tilde{B}_i$, $i = 1, 2, \dots, r$
--

$\underbrace{\hspace{15em}}_{\text{antecedent}} \qquad \underbrace{\hspace{5em}}_{\text{consequent}}$

- $x_1, x_2, \dots$ are the fuzzy/linguistic variables

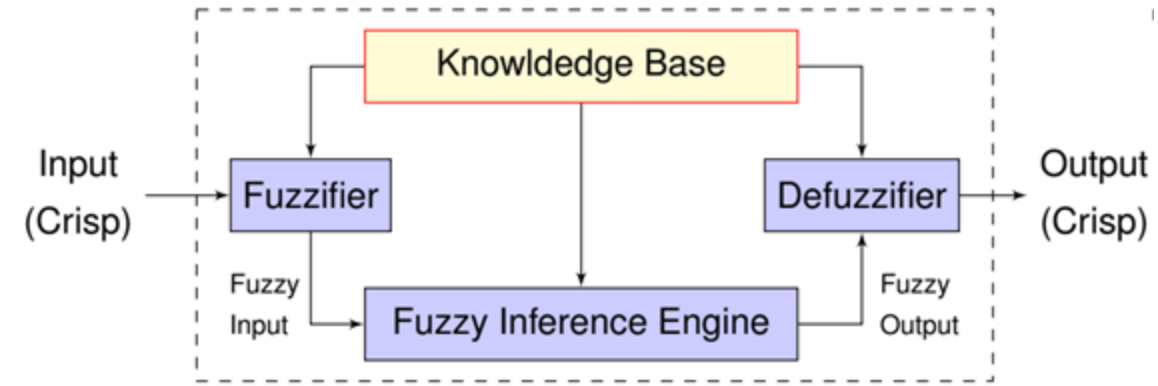
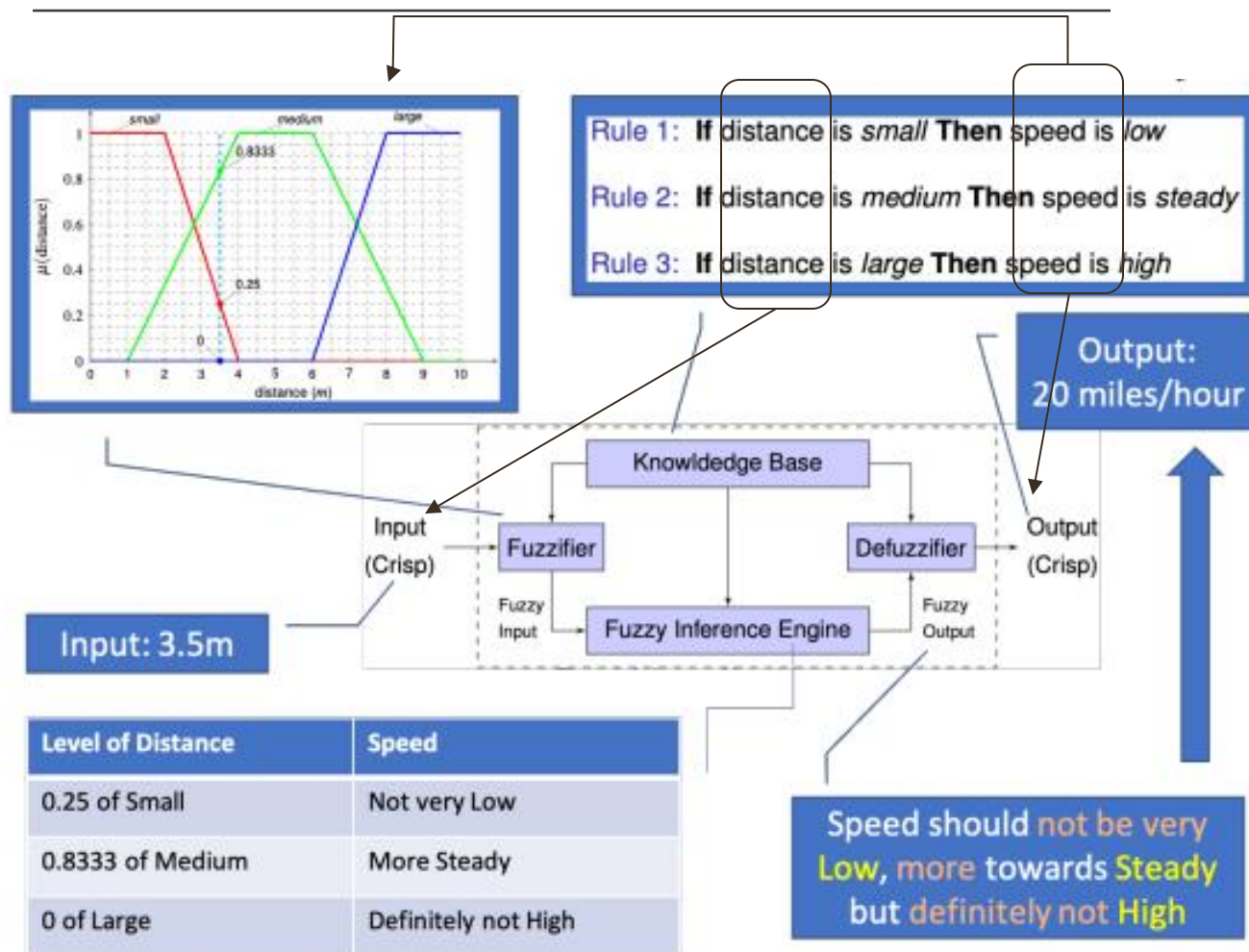
Knowledge Base (2)

General rule format:

$$\text{Rule } i: \text{IF } \underbrace{x_1 \text{ is } \tilde{A}_{i1} \text{ and/or } x_2 \text{ is } \tilde{A}_{i2} \text{ and/or } \dots}_{\text{antecedent}} \text{ THEN } \underbrace{y \text{ is } \tilde{B}_i}_{\text{consequent}}, i = 1, 2, \dots, r$$

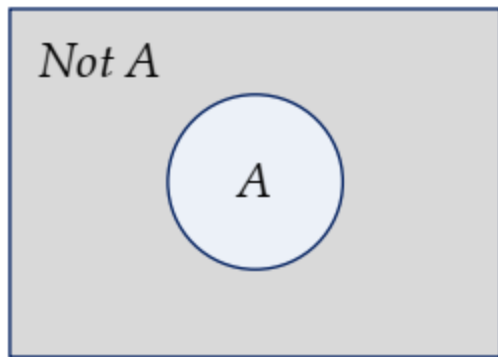
- y is the output of the fuzzy inference system.
- “and” and “or” are fuzzy operators
- $\tilde{A}_{i1}, \tilde{A}_{i2}, \dots$ are the fuzzy sets (associated with a linguistic variable) representing the i th antecedent pairs
- $\tilde{B}_i$ is the fuzzy set (associated with a linguistic variable) representing the i th consequent
- r is number of rules.

Knowledge Base (3)

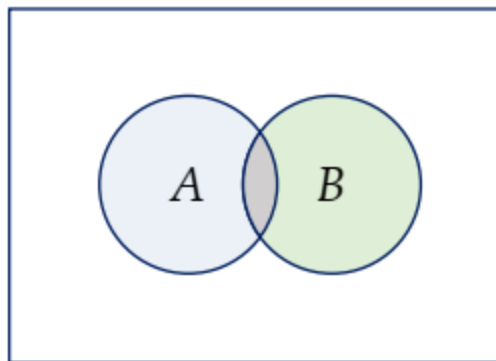


Fuzzy Inference Engine

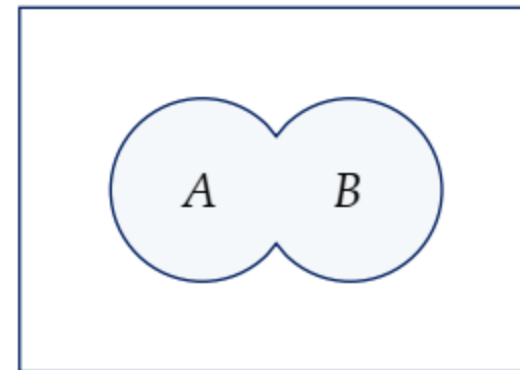
- Fuzzy inference engine is to produce the fuzzy output according to the crisp inputs based on the knowledge (knowledge base) represented by IF-THEN rule. This is the process of reasoning. It generally involves two processes, i.e., **rule evaluation and rule aggregation**.
 - Rule evaluation (implication) is to **apply the fuzzy set operators (AND, OR, NOT) to the antecedents** to determine the firing strength of each rule.
 - Rule aggregation is to combine the output (consequents) fuzzy sets using the firing strengths obtained in the process of rule evaluation.
- There are three standard fuzzy set operations:
 - Fuzzy union operation (OR), also known as t-norm or conjunction operator.
 - Fuzzy intersection operation (AND), also known as t-conorm, s-norm operation, disjunction operation.
 - Fuzzy complement operation (NOT)



Complement



Intersection



Union

Rule Evaluation Example

Example - Linguistic Rules (max for “or” and min for “and”; $x = 3.5$, $y = 1$)

Rule 1: If x is *small* **and** y is *negative* **Then** z is *low*

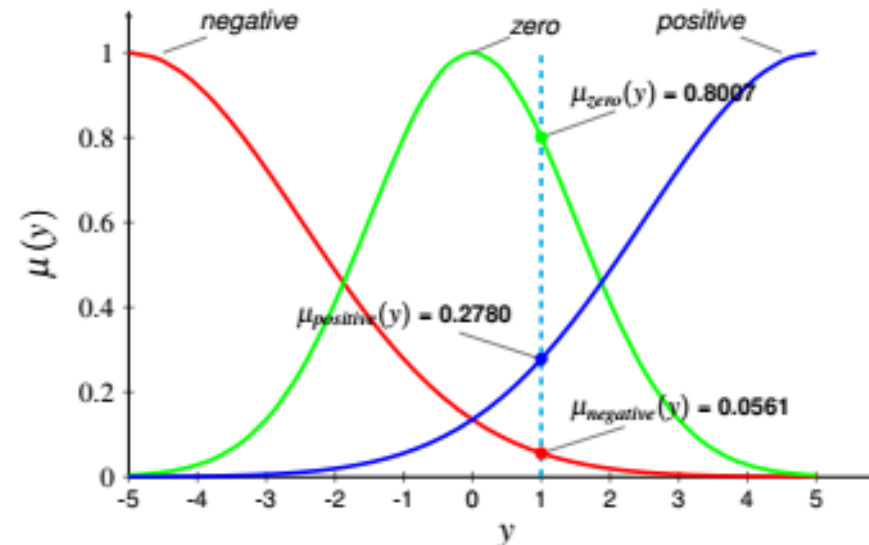
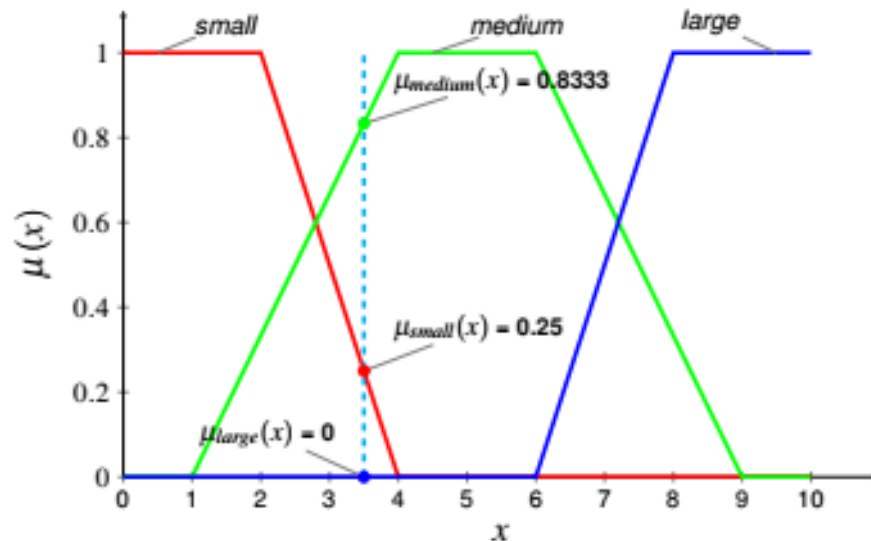
(Rule evaluation: $\min(\mu_{\text{small}}(x), \mu_{\text{negative}}(y)) = \min(0.25, 0.0561) = 0.0561$)

Rule 2: If x is *medium* **or** y is *zero* **Then** z is *middle*

(Rule evaluation: $\max(\mu_{\text{medium}}(x), \mu_{\text{zero}}(y)) = \max(0.833, 0.8007) = 0.833$)

Rule 3: If x is *large* **or** y is *not positive* **Then** z is *high*

(Rule evaluation: $\max(\mu_{\text{large}}(x), \mu_{\text{not positive}}(y)) = \max(0, 1 - 0.2780) = 0.7220$)



Many kinds of rule evaluation based on rule operator

Fuzzy OR operator (fuzzy union operator)

- *Maximum:* $\mu_{\underline{A} \cup \underline{B}}(x, y) = \mu_{\underline{A}}(x) \vee \mu_{\underline{B}}(y) = \max(\mu_{\underline{A}}(x), \mu_{\underline{B}}(y))$
- *Algebraic sum:* $\mu_{\underline{A} \cup \underline{B}}(x, y) = \mu_{\underline{A}}(x) + \mu_{\underline{B}}(y) - \mu_{\underline{A}}(x) \times \mu_{\underline{B}}(y)$

Fuzzy AND operator (fuzzy intersection operator)

- *Minimum:* $\mu_{\underline{A} \cap \underline{B}}(x, y) = \mu_{\underline{A}}(x) \wedge \mu_{\underline{B}}(y) = \min(\mu_{\underline{A}}(x), \mu_{\underline{B}}(y))$
- *Product:* $\mu_{\underline{A} \cap \underline{B}}(x, y) = \mu_{\underline{A}}(x) \times \mu_{\underline{B}}(y)$

Fuzzy NOT operator (fuzzy complement operator)

- *Complement:* $\mu_{\underline{A}^-}(x) = 1 - \mu_{\underline{A}}(x)$

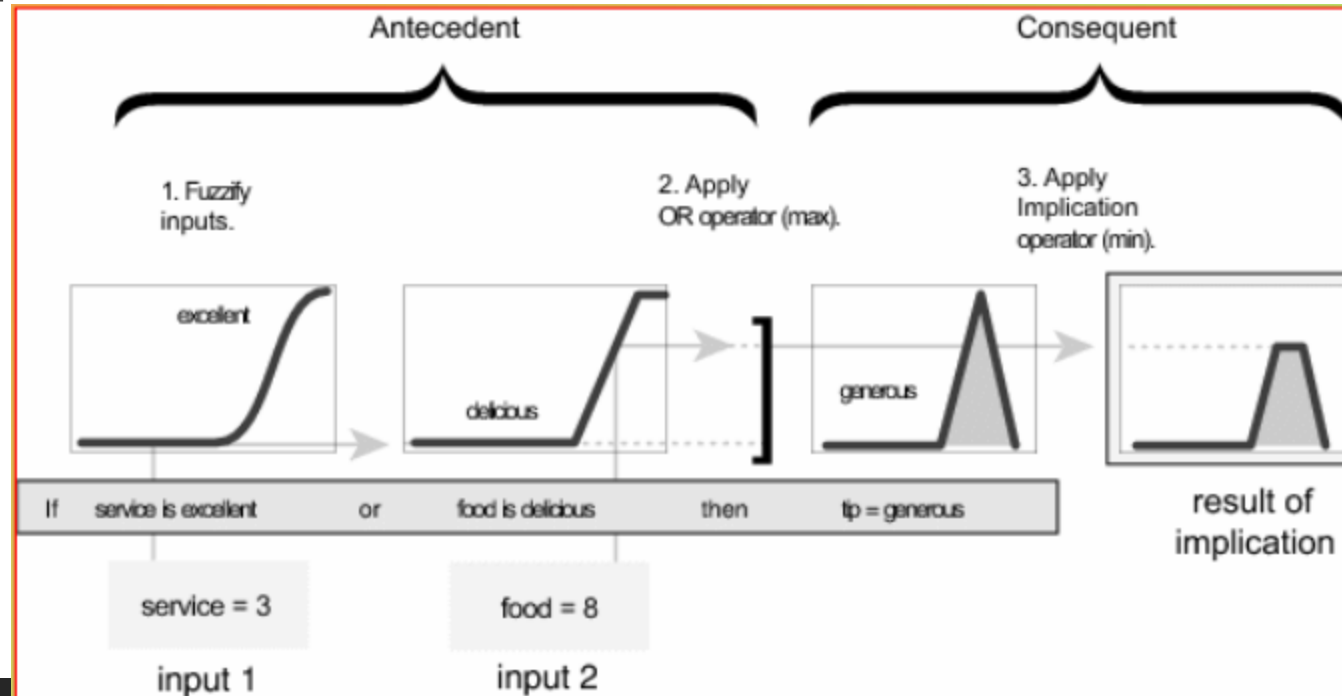
Common Fuzzy Inference

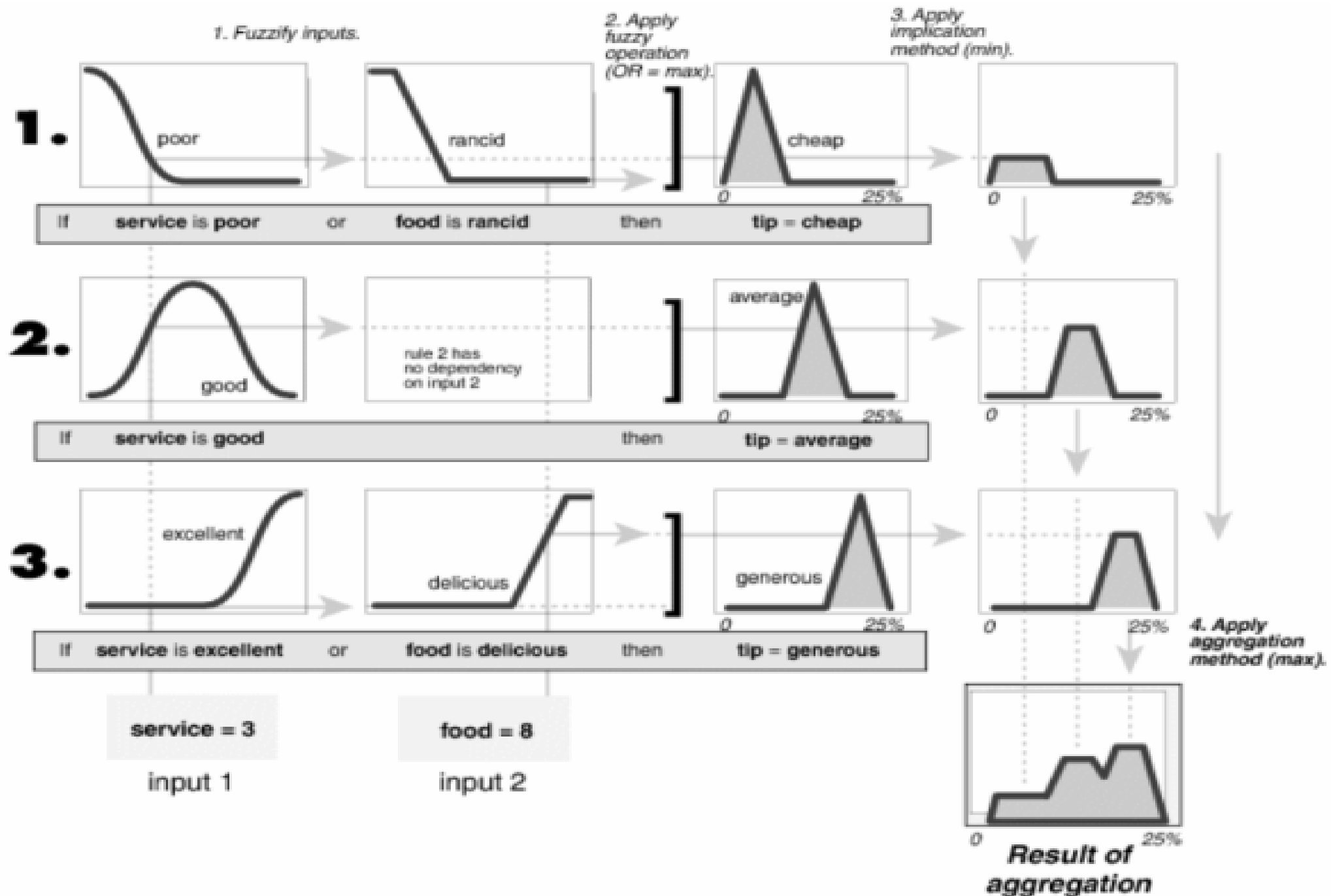
- Mamdani
- Sugeno
- Tsukamoto

Mamdani System

About Mamdani

Mamdani-type inference, **expects the output membership functions to be fuzzy sets**. After the aggregation process, there is a fuzzy set for each output variable that needs defuzzification.





Mamdani Systems

IF x_1 is A_1 and x_2 is A_2 THEN y is B

IF x_1 is A_1^k and x_2 is A_2^k THEN y^k is B^k , where $k = 1, 2, \dots, r$

IF x_1 is A_1 or x_2 is A_2 THEN y is B

IF x_1 is A_1^k or x_2 is A_2^k THEN y^k is B^k , where $k = 1, 2, \dots, r$

**

AND operator: Take minimum of the membership values

OR operator: Take maximum of the membership values

Cases:

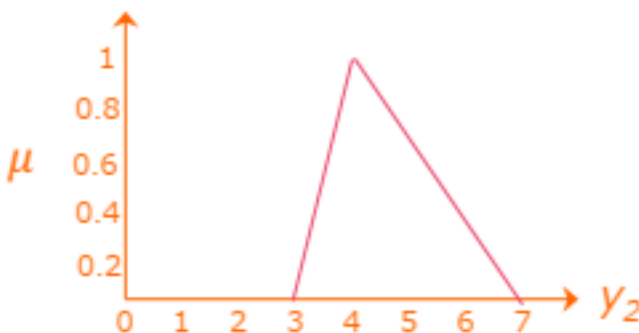
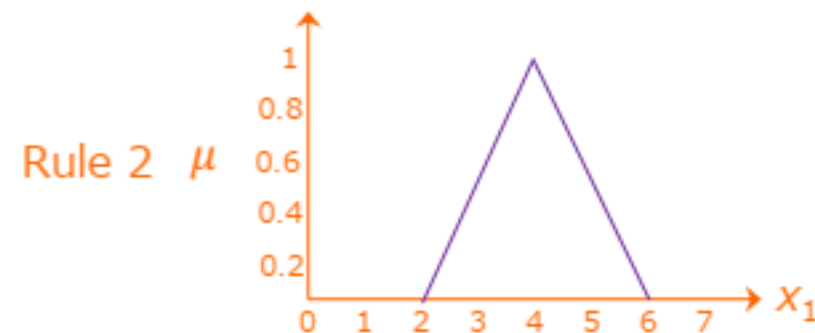
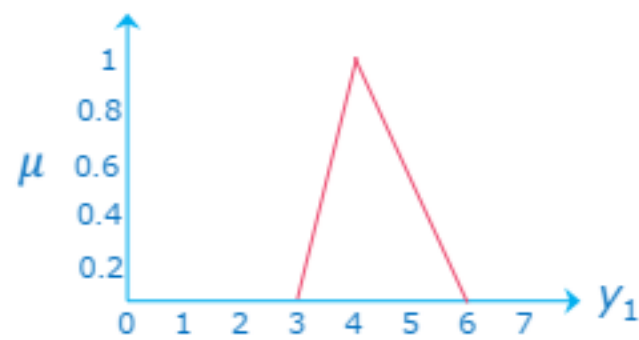
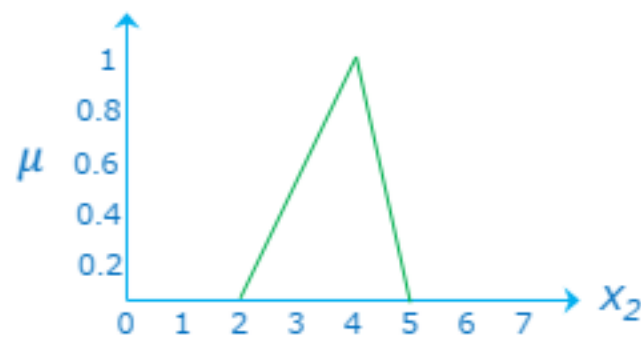
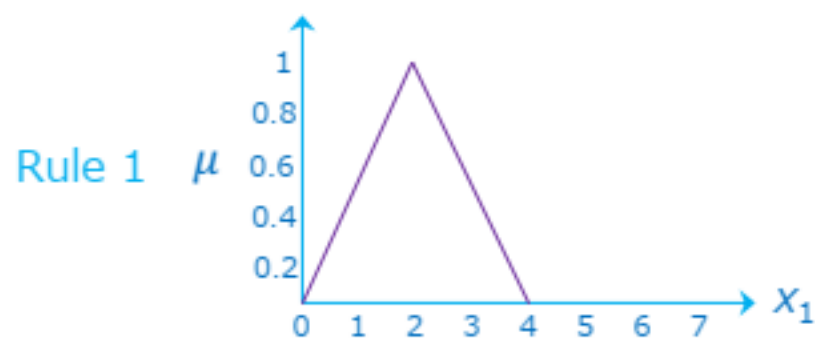
1. Max-min inference method
2. Max-product inference method

Min-max Inference Method

Consider a sample 2 rule system where each rule has 2 antecedents and one consequent:

Rule 1: IF x_1 is A_1^1 and x_2 is A_2^1 THEN y^1 is B^1

Rule 2: IF x_1 is A_1^2 or x_2 is A_2^2 THEN y^2 is B^2



Min-max Inference Method

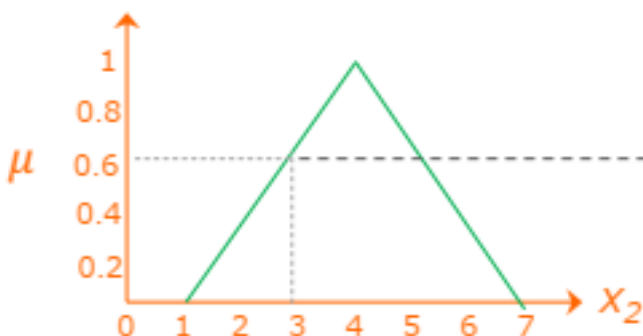
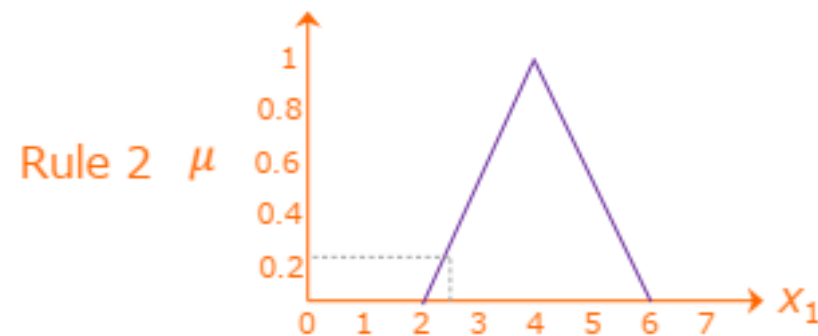
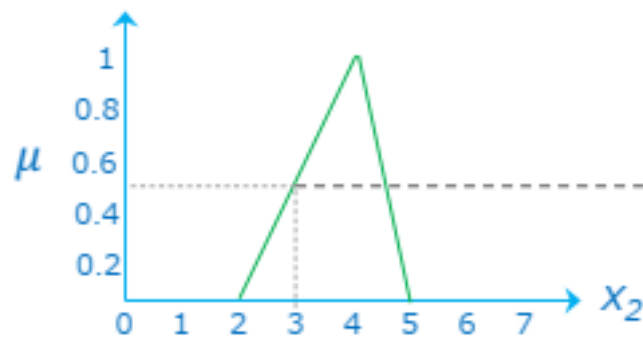
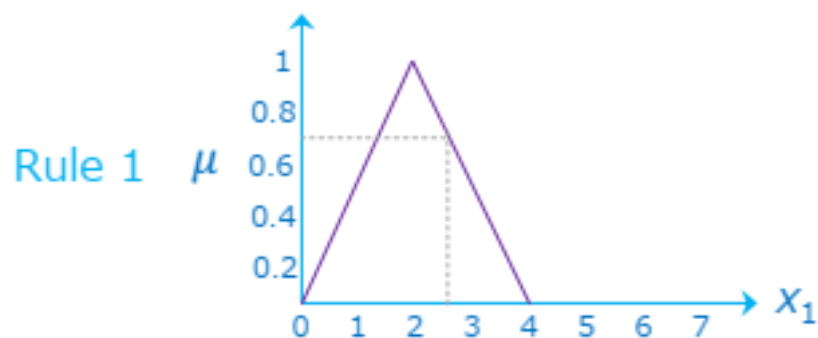
Consider a sample 2 rule system where each rule has 2 antecedents and one consequent:

Rule 1: IF x_1 is A_1^1 and x_2 is A_2^1 THEN y_1 is B_1

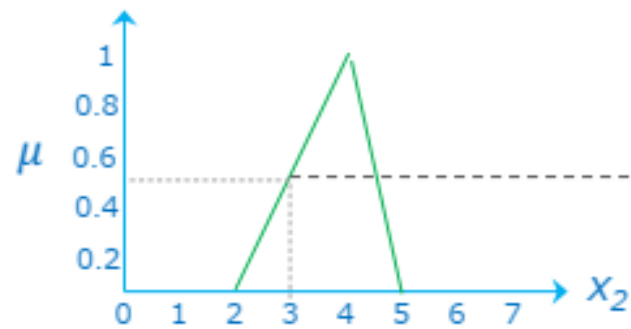
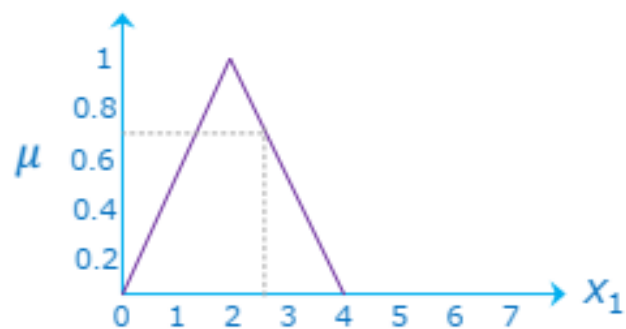
Rule 2: IF x_1 is A_1^2 or x_2 is A_2^2 THEN y_2 is B_2

$$x_1 = 2.5$$

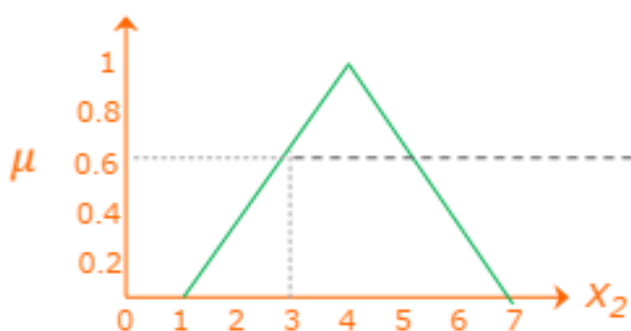
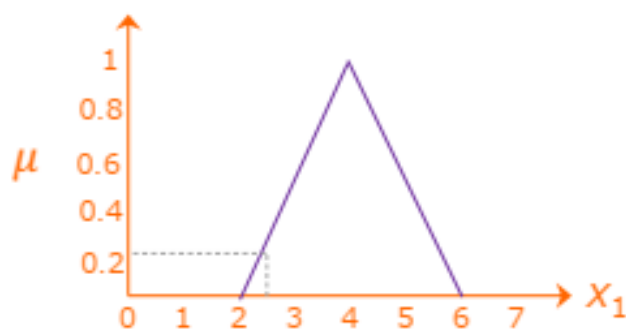
$$x_2 = 3$$



Rule 1



Rule 2



Truncated membership function

Min-product Inference Method

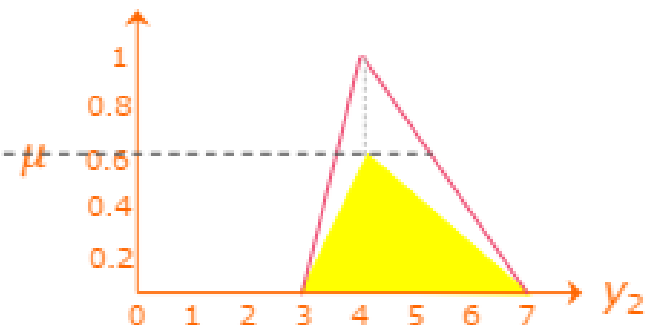
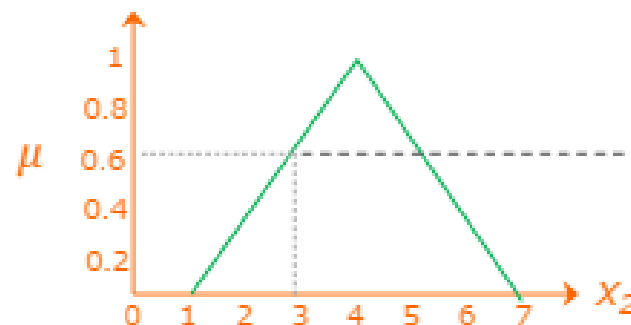
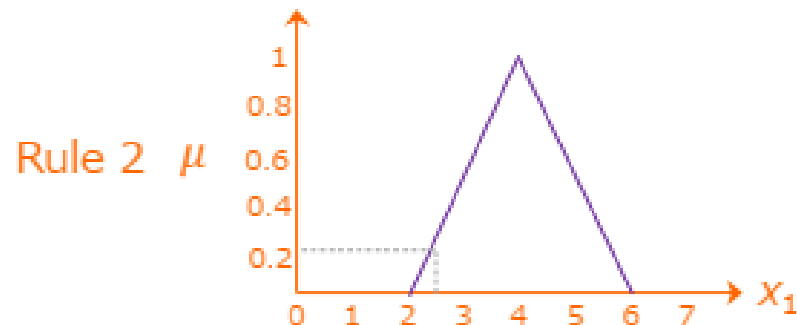
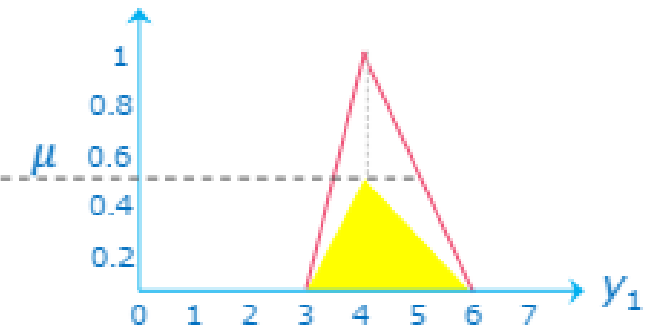
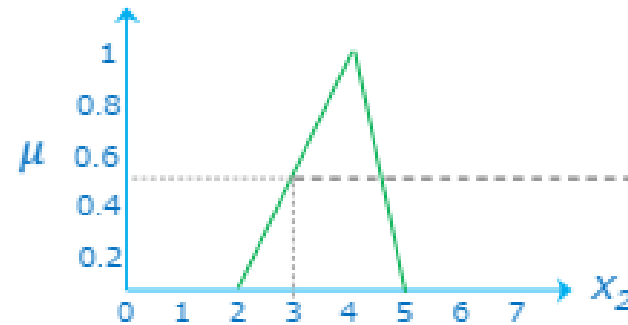
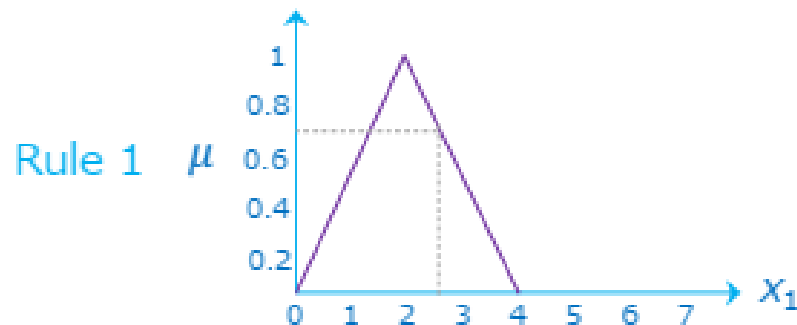
Consider a sample 2 rule system where each rule has 2 antecedents and one consequent:

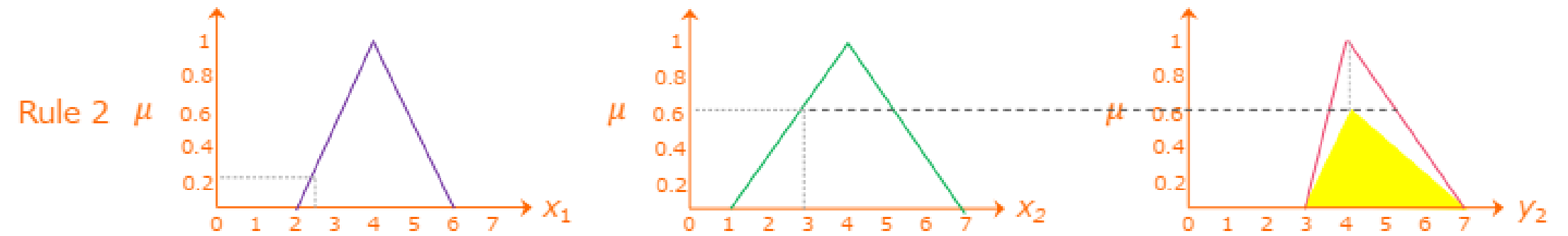
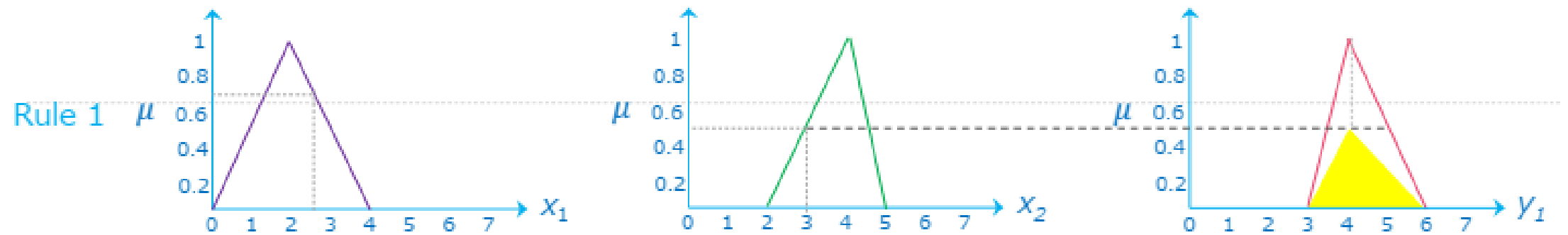
Rule 1: IF x_1 is A_1^1 and x_2 is A_2^1 THEN y_1 is B_1

$$x_1 = 2.5$$

$$x_2 = 3$$

Rule 2: IF x_1 is A_1^2 or x_2 is A_2^2 THEN y_2 is B_2





Scaled membership function

Sugeno System

About Sugeno System

In Sugeno , the first two parts of the fuzzy inference process, **fuzzifying the inputs and applying the fuzzy operator**, are exactly the same with Mamdani. The main difference between Mamdani and Sugeno is that the **Sugeno output membership functions are either linear or constant and usually the output is in an operation.**

$$\left\{ \begin{array}{l} \text{If } X \text{ is small and } Y \text{ is small then } z = -x + y + 1. \\ \text{If } X \text{ is small and } Y \text{ is large then } z = -y + 3. \\ \text{If } X \text{ is large and } Y \text{ is small then } z = -x + 3. \\ \text{If } X \text{ is large and } Y \text{ is large then } z = x + y + 2. \end{array} \right.$$

Sugeno Systems

IF x_1 is A_1 and x_2 is A_2 THEN y is $y=f(x_1, x_2)$

IF x_1 is A_1 or x_2 is A_2 THEN y is $y=f(x_1, x_2)$

where $y=f(x_1, x_2)$ is a crisp function in the consequent.

The overall aggregated output will be obtained through weighted average defuzzification method:

$$y^* = \frac{\sum_{i=1}^r (\mu_{y_i} \cdot y_i)}{\sum_{i=1}^r \mu_{y_i}}$$

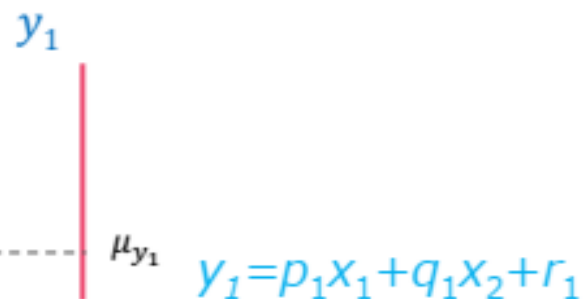
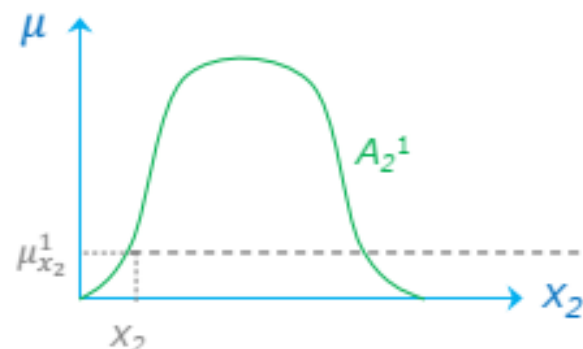
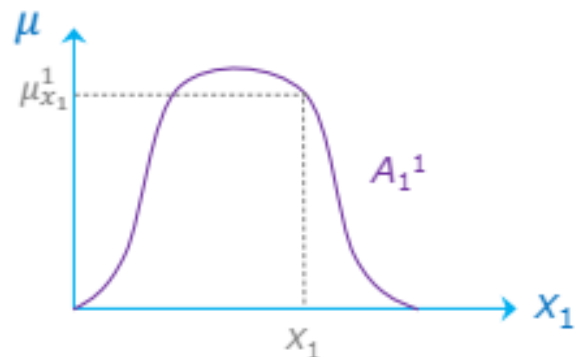
Sugeno Systems

Consider a sample 2 rule system where each rule has 2 antecedents and one consequent:

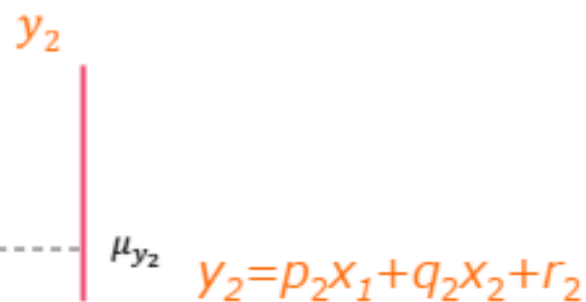
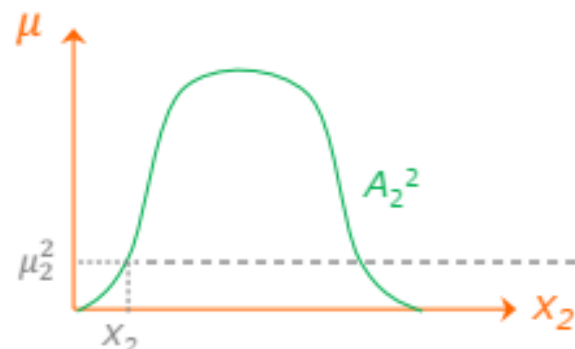
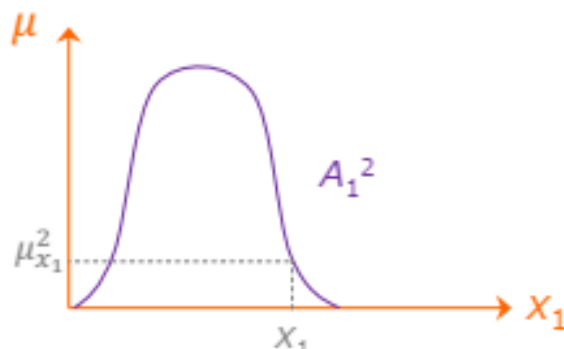
Rule 1: IF x_1 is A_1^1 and x_2 is A_2^1 THEN y_1 is $y_1 = p_1x_1 + q_1x_2 + r_1$

Rule 2: IF x_1 is A_1^2 and x_2 is A_2^2 THEN y_2 is $y_2 = p_2x_1 + q_2x_2 + r_2$

Rule 1



Rule 2



Sugeno Systems

$$y^* = \frac{\sum_{i=1}^r (\mu_{y_i} \cdot y_i)}{\sum_{i=1}^r \mu_{y_i}}$$

Consider a two input – single output Sugeno model with 4 rules as:

Rule 1: IF x_1 is small and x_2 is small, THEN $y_1 = (-x_1) + x_2 + 1$

Rule 2: IF x_1 is small and x_2 is large, THEN $y_2 = (-x_2) + 3$

Rule 3: IF x_1 is large and x_2 is small, THEN $y_3 = (-x_1) + 3$

Rule 4: IF x_1 is large and x_2 is large, THEN $y_4 = (-x_1) + x_2 + 2$

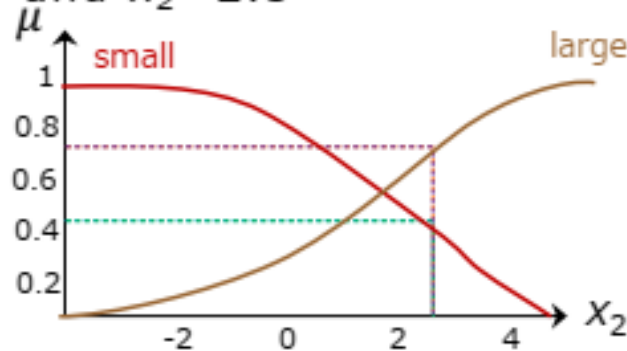
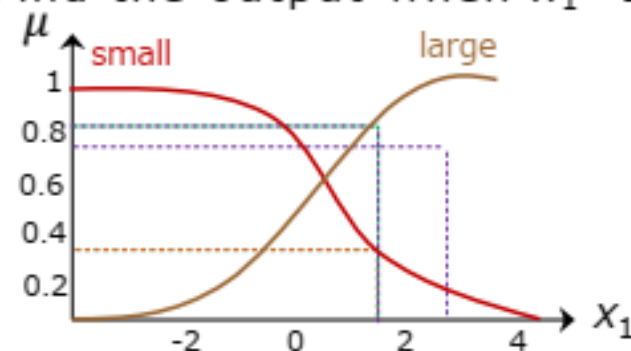
$$y_1 = (-1.5) + 2.5 + 1 = 2$$

$$y_2 = (-2.5) + 3 = 0.5$$

$$y_3 = (-1.5) + 3 = 1.5$$

$$y_4 = (-1.5) + 2.5 + 2 = 3$$

Find the output when $x_1 = 1.5$ and $x_2 = 2.5$



$$\mu_{y_r} = \min(\mu_{x_1}, \mu_{x_2})$$

$$\mu_{y_1} = \min(0.8, 0.3) = 0.3$$

$$\mu_{y_2} = \min(0.8, 0.7) = 0.7$$

$$\mu_{y_3} = \min(0.2, 0.3) = 0.2$$

$$\mu_{y_4} = \min(0.2, 0.7) = 0.2$$

$$y^* = \frac{0.3 \times 2 + 0.7 \times 0.5 + 0.2 \times 1.5 + 0.2 \times 3}{0.3 + 0.7 + 0.2 + 0.2} = \frac{3.45}{1.7} = 2.03$$

Mamdani or Sugeno?

MAMDANI

- Widely accepted for capturing expert knowledge.
- Allows us to describe the expertise in more intuitive, more human-like manner.
- Entails a substantial computational burden.

SUGENO

- Computationally effective and works well with optimization and adaptive techniques, which makes it very attractive in control problems, particularly for dynamic nonlinear systems.

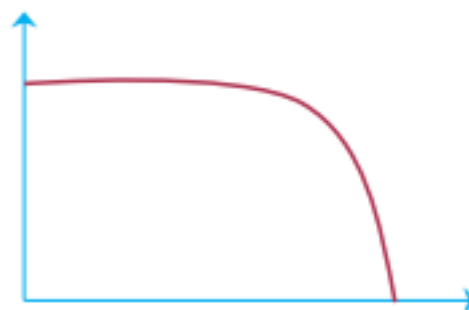
Tsukamoto Systems

Tsukamoto Systems

IF x_1 is A_1 and x_2 is A_2 THEN y is B

IF x_1 is A_1 or x_2 is A_2 THEN y is B

where the fuzzy set B has a monotonic membership. Monotonic function (shoulder function) is a function whose successive values are increasing, decreasing, or constant.



The overall aggregated output will be obtained through weighted average defuzzification method:

$$y^* = \frac{\sum_{i=1}^r (\mu_{y_i} \cdot y_i)}{\sum_{i=1}^r \mu_{y_i}}$$

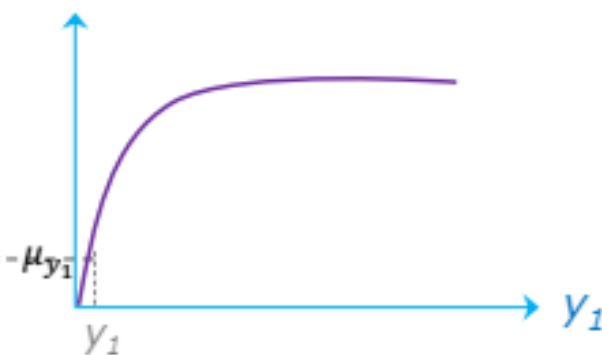
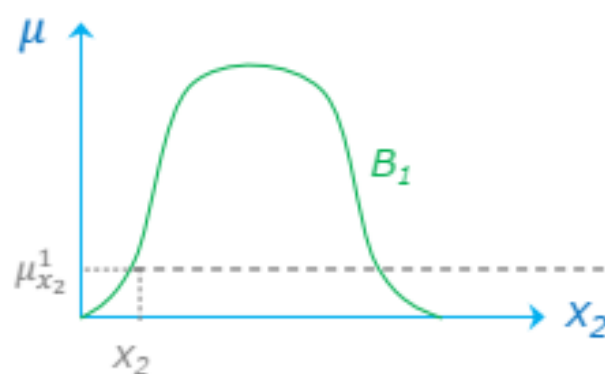
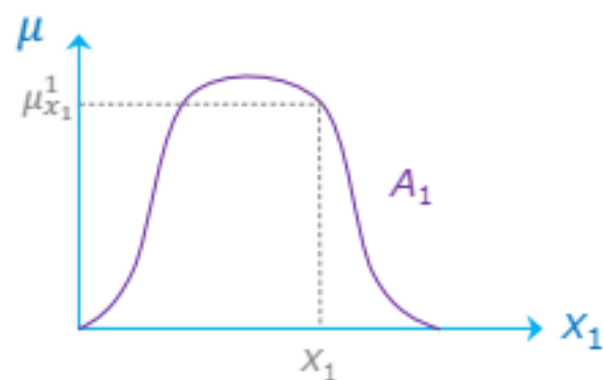
Tsukamoto Systems

The inferred output of each rule is defined as a crisp value induced by the membership value coming from the antecedent of the rule.

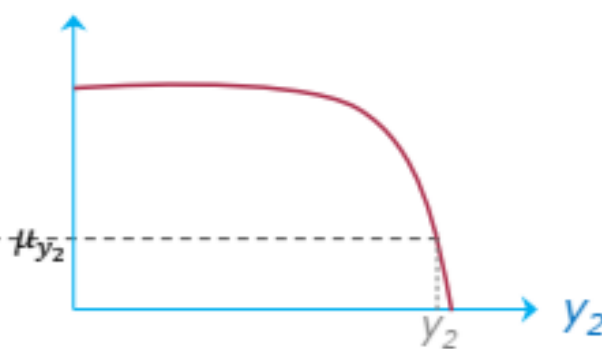
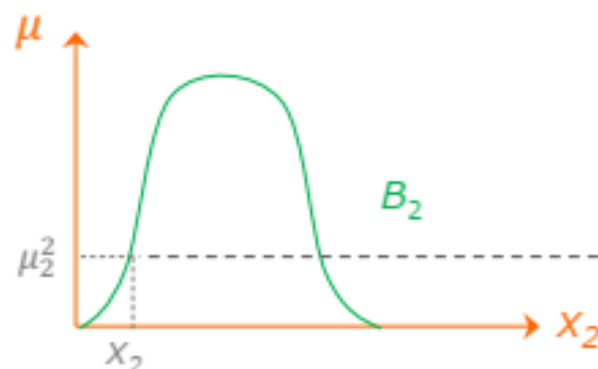
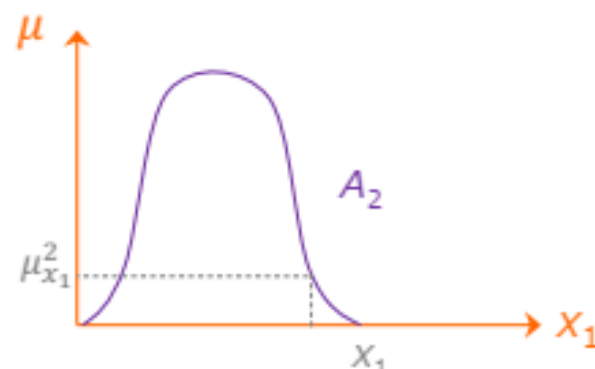
Rule 1: IF x_1 is A_1 and x_2 is B_1 THEN y_1 is C_1

Rule 2: IF x_1 is A_2 or x_2 is B_2 THEN y_2 is C_2

Rule 1



Rule 2



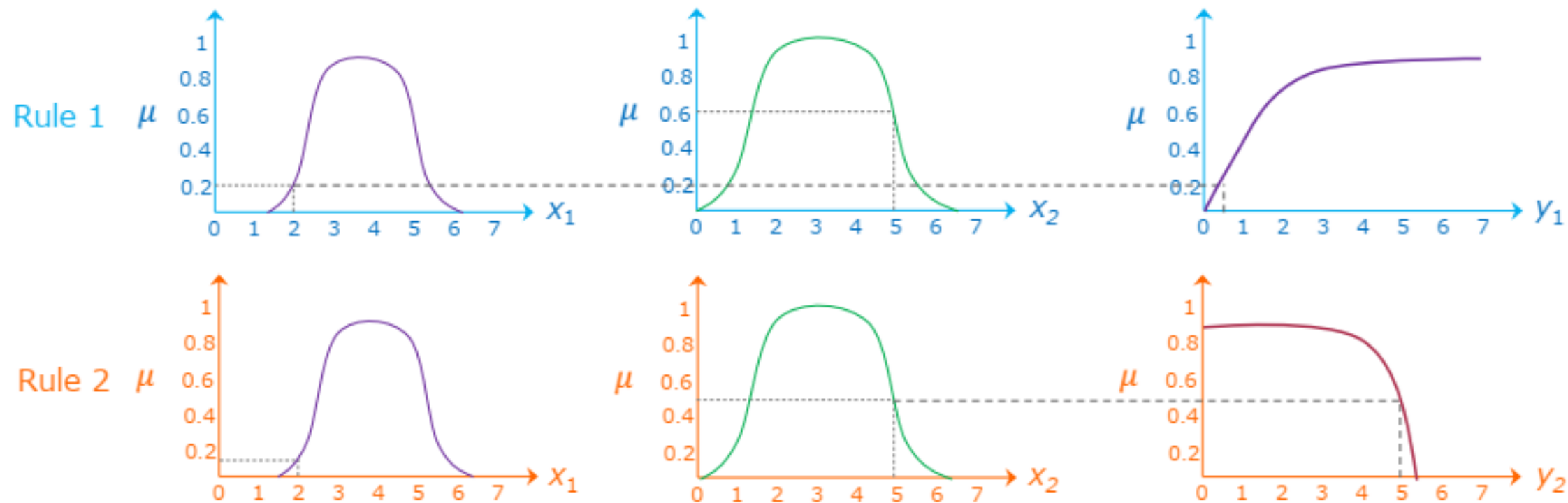
Tsukamoto Systems

Rule 1: IF x_1 is A_1 and x_2 is B_1 THEN y_1 is C_1

Rule 2: IF x_1 is A_2 or x_2 is B_2 THEN y_2 is C_2

$$x_1 = 2$$

$$x_2 = 5$$



$$y^* = \frac{\mu_{y_1} \times y_1 + \mu_{y_2} \times y_2}{\mu_{y_1} + \mu_{y_2}} = \frac{0.2 \times 0.5 + 0.6 \times 5}{0.2 + 0.6} = 3.71$$

Tsukamoto Model

- Advantage

Tsukamoto model avoids time consuming process of defuzzification since each rule infers a crisp output and the overall output can be calculated using weighted average method.

- Disadvantage

Since the output membership function should be monotonic in nature, it is not as useful as a general approach. Hence, it can only be used in specific situations.

Example of the Differences#

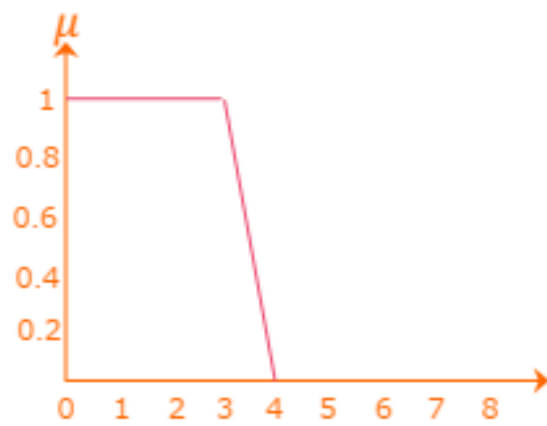
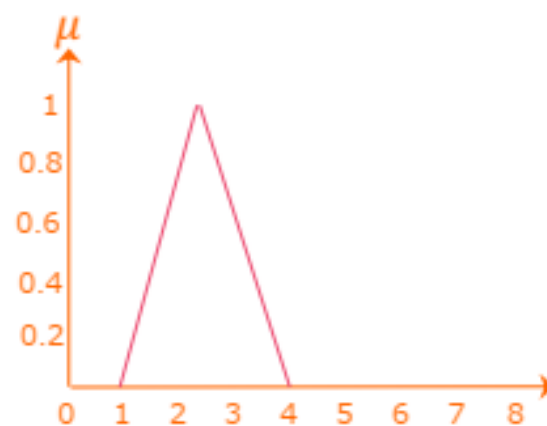
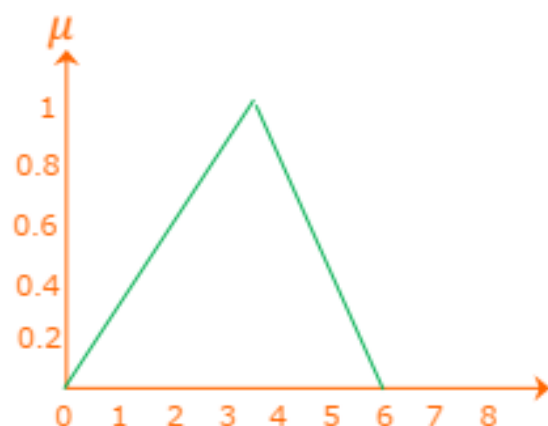
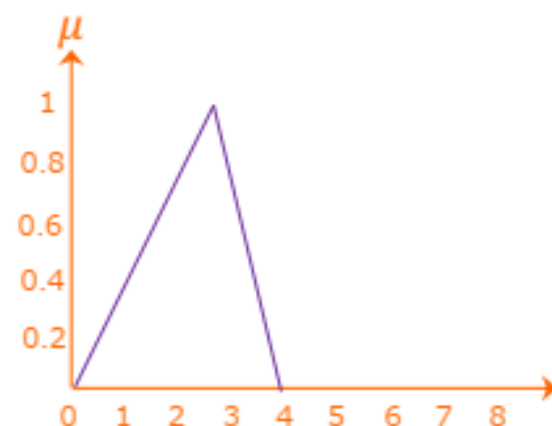
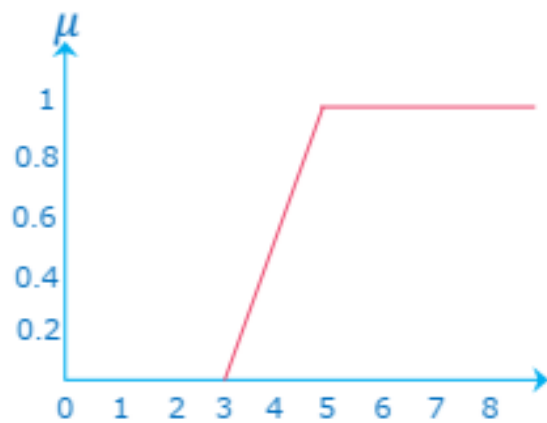
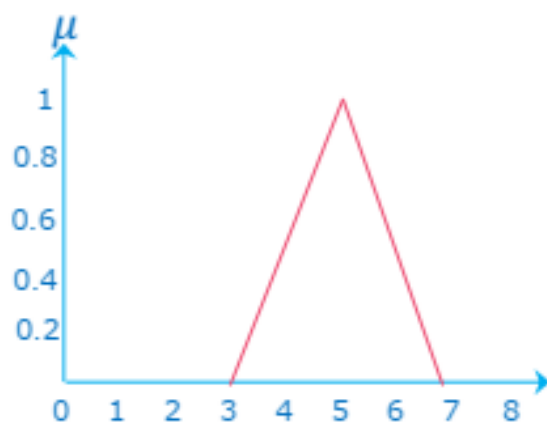
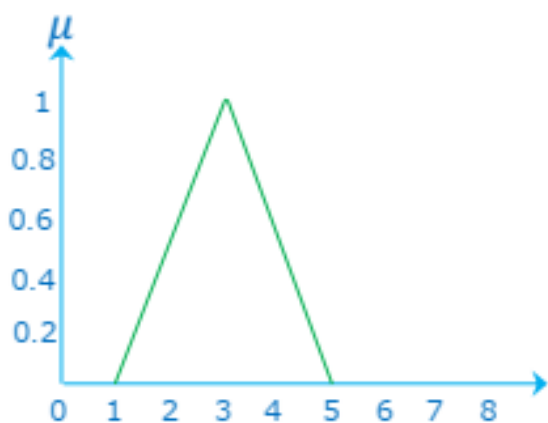
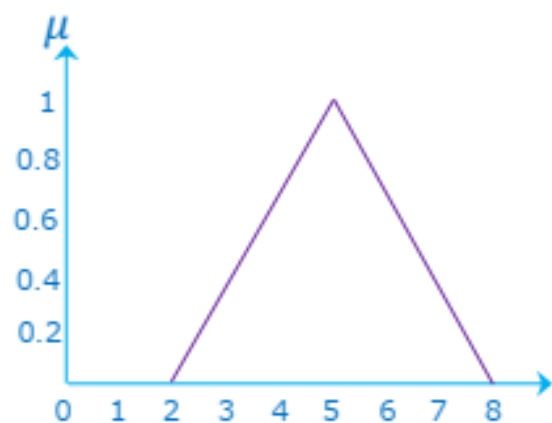
Consider a system with the following rules:

Rule 1: IF x is A_1 and y is B_1 THEN z_1 is C_1 $\Rightarrow z_1 = x + 3y - 6$

Rule 2: IF x is A_2 or y is B_2 THEN z_2 is C_2 $\Rightarrow z_2 = 2x + y - 7$

$$x = 3$$

$$y = 4$$



Consider a system with the following rules:

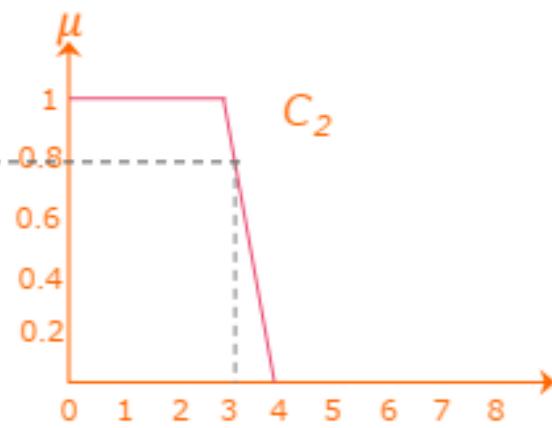
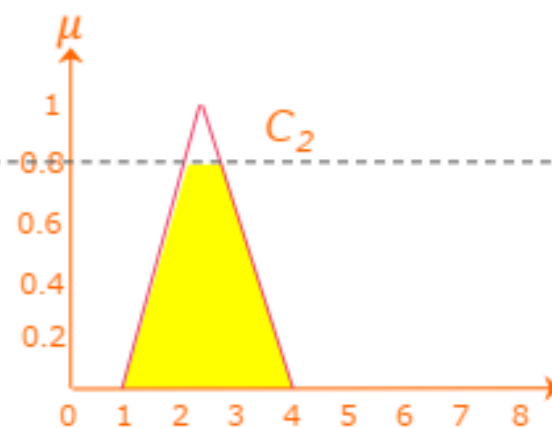
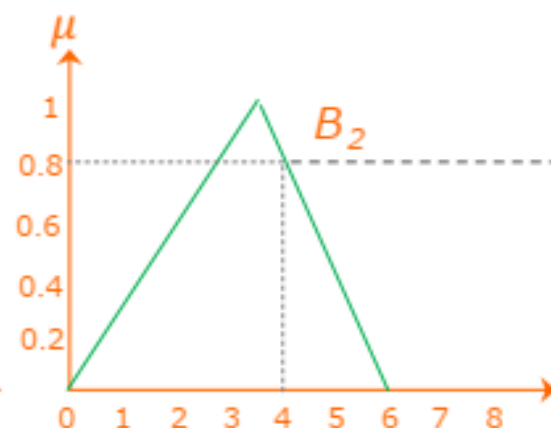
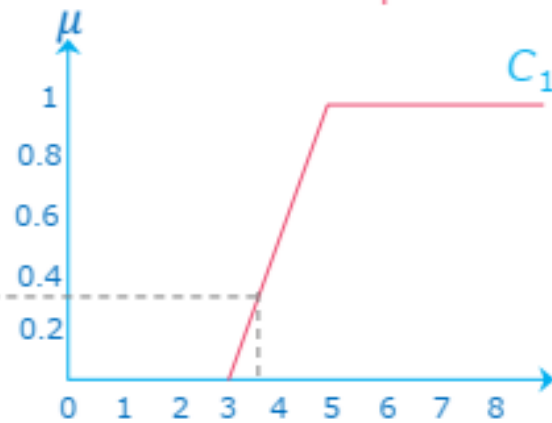
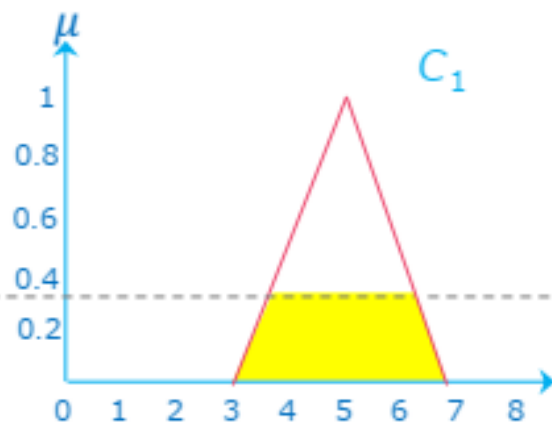
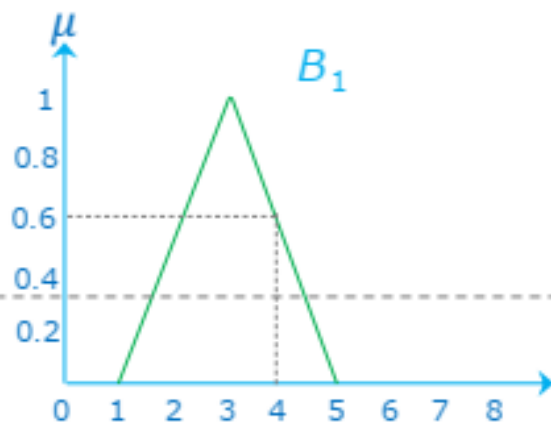
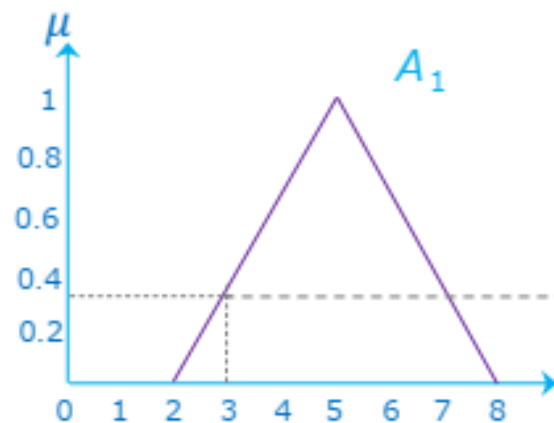
Rule 1: IF x is A_1 and y is B_1 THEN z_1 is C_1 $\Rightarrow z_1 = x + 3y - 6$

Rule 2: IF x is A_2 or y is B_2 THEN z_2 is C_2 $\Rightarrow z_2 = 2x + y - 7$

$$x = 3$$

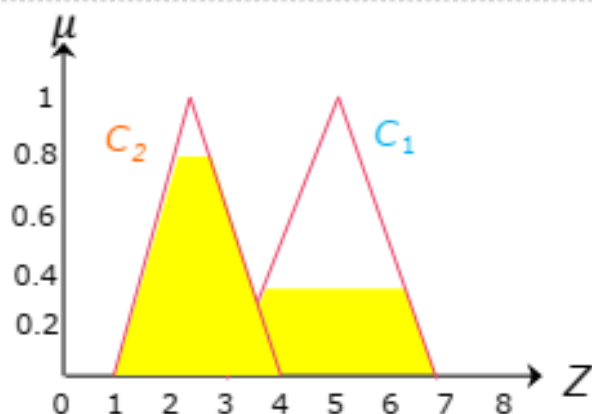
$$y = 4$$

Monotonic Membership Function



Mamdani vs. Sugeno vs. Tsukamoto

M
a
m
d
a
n
i



$$Z^* = \frac{\mu_1 \bar{z}_1 + \mu_2 \bar{z}_2}{\mu_1 + \mu_2}$$

$$\mu_1 = 0.3$$

$$\mu_2 = 0.8$$

μ_i is the membership value of z_i

$\bar{z}_i$ is the center of gravity of z_i

$$\bar{z}_1 = 5$$

$$\bar{z}_2 = 2.5$$

$$Z^* = \frac{0.3 \times 5 + 0.8 \times 2.5}{0.3 + 0.8} = 3.181$$

S
u
g
e
n
o

$$z_1 = x + 3y - 6$$

$$x = 3$$

$$y = 4$$

$$z_2 = 2x + y - 7$$

$$z_1 = 3 + 3 \times 4 - 6 = 9 \quad z_2 = 2 \times 3 + 4 - 7 = 3$$

$$\mu_1 = 0.3$$

$$\mu_2 = 0.8$$

$$Z^* = \frac{0.3 \times 9 + 0.8 \times 3}{0.3 + 0.8} = 4.636$$

T
s
u
k
a
m
o
t
o

$$\mu_1 = 0.3$$

$$\mu_2 = 0.8$$

$$\bar{z}_1 = 3.5$$

$$\bar{z}_2 = 3$$

$$Z^* = \frac{0.3 \times 3.5 + 0.8 \times 3}{0.3 + 0.8} = 3.136$$

Example #1

Example #1 (1/13)

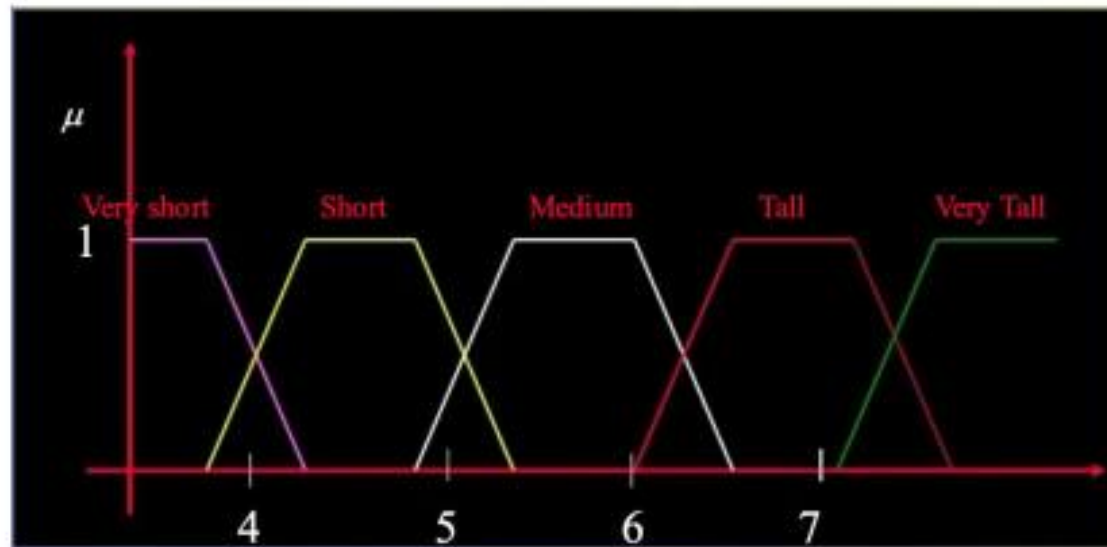
We want to evaluate the health of a person based on his height and weight

- The input variables are the crisp numbers of the person's height and weight
- Fuzzification is a process by which the numbers are changes into linguistic words

Example #1 (2/13)

Step 1: Fuzzification

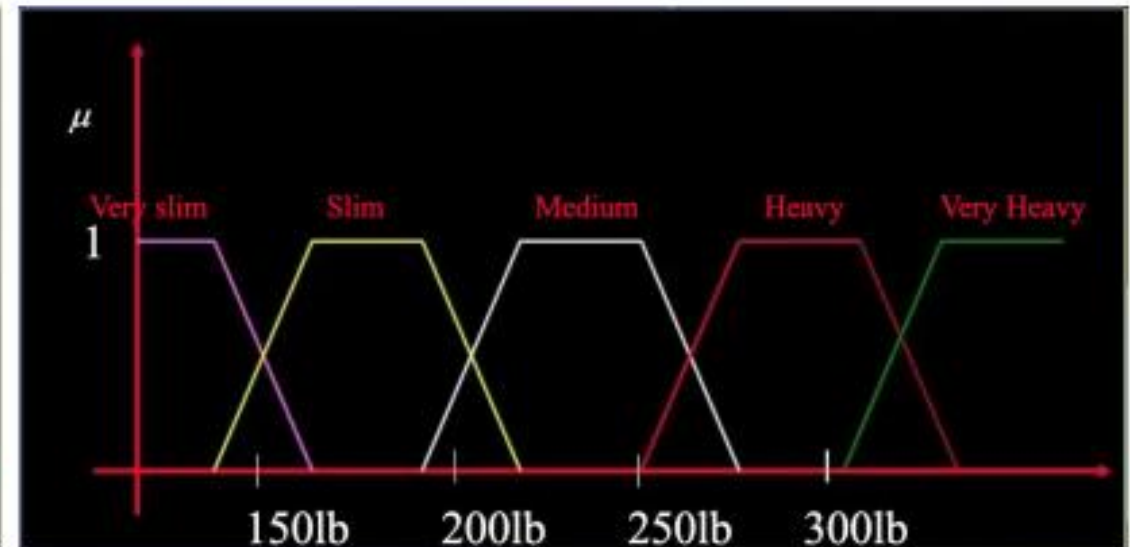
Fuzzification of height



VS = Very short
S = Short
M = Medium

T = Tall
VT = Very Tall

Fuzzification of weight



VS = Very slim
S = Slim
M = Medium

H = Heavy
VH = Very Heavy

Example #1 (3/13)

Step 2: Rules

- Rules reflect experts' decisions
- Rules are tabulated as fuzzy words
- Rules can be grouped in subsets
- Rules can be redundant
- Rules can be adjusted to match desired results

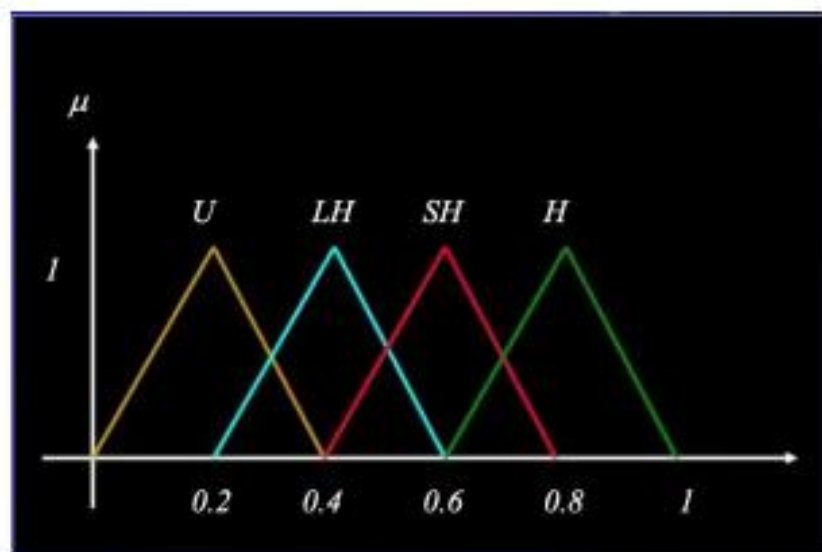
Example #1 (4/13)

Step 2: Rules

- Healthy (H)
- Somewhat healthy (SH)
- Less Healthy (LH)
- Unhealthy (U)

} Rule function
 $f = \{U, LH, SH, H\}$

Fuzzified Decision



Fuzzy Rules Table

		Weight				
Height		Very Slim	Slim	Medium	Heavy	Very Heavy
	Very Short	H	SH	LH	U	U
	Short	SH	H	SH	LH	U
	Medium	LH	H	H	LH	U
	Tall	U	SH	H	SH	U
	Very Tall	U	LH	H	SH	LH

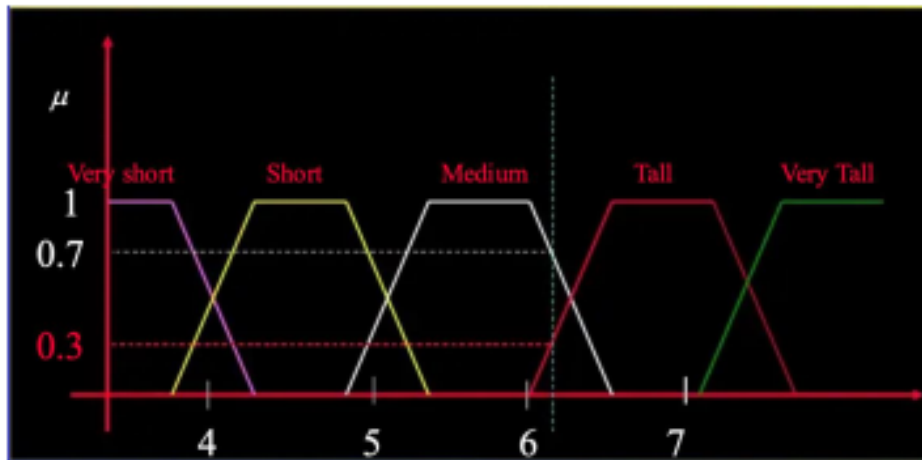
Example #1 (5/13)

Step 3: Calculate

For a given person, compute the membership of his/her weight & height

- Assume that a person height is 6.1'
- Assume that the person's weight is 140 lb

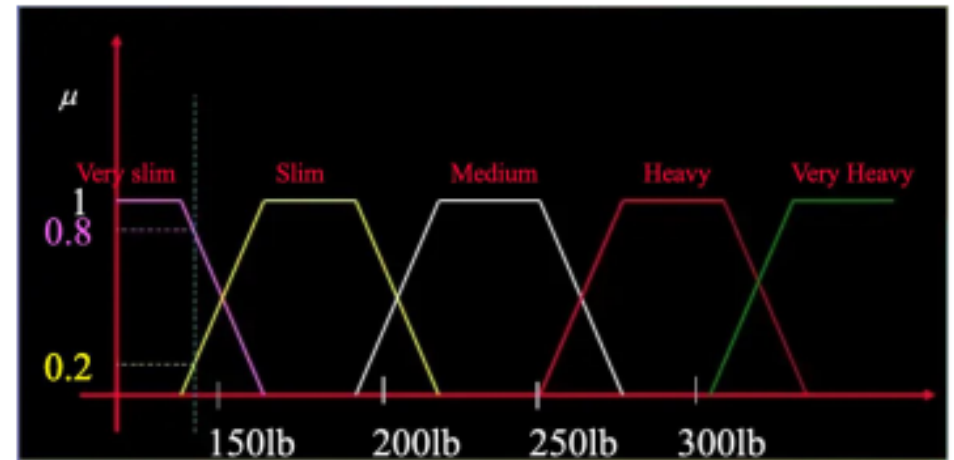
Membership of Height



$$\mu_{\text{height}} = \{\mu_{VS} \quad \mu_S \quad \mu_M \quad \mu_T \quad \mu_{VT}\}$$

$$\mu_{\text{height}} = \{0 \quad 0 \quad 0.7 \quad 0.3 \quad 0\}$$

Membership of Weight



$$\mu_{\text{weight}} = \{\mu_{VS} \quad \mu_S \quad \mu_M \quad \mu_H \quad \mu_{VH}\}$$

$$\mu_{\text{weight}} = \{0.8 \quad 0.2 \quad 0 \quad 0 \quad 0\}$$

Example #1 (6/13)

Step 4: Activate Rules

$$\mu_{height} = \{\mu_{VS} \quad \mu_S \quad \mu_M \quad \mu_T \quad \mu_{VT}\}$$

$$\mu_{height} = \{0 \quad 0 \quad 0.7 \quad 0.3 \quad 0\}$$

$$\mu_{weight} = \{\mu_{VS} \quad \mu_S \quad \mu_M \quad \mu_H \quad \mu_{VH}\}$$

$$\mu_{weight} = \{0.8 \quad 0.2 \quad 0 \quad 0 \quad 0\}$$

Fuzzy Rules Table

		Weight				
Height		Very Slim	Slim	Medium	Heavy	Very Heavy
	Very Short	H	SH	LH	U	U
	Short	SH	H	SH	LH	U
	Medium	LH	H	H	LH	U
	Tall	U	SH	H	SH	U
	Very Tall	U	LH	H	SH	LH



		Weight				
Height		Very Slim	Slim	Medium	Heavy	Very Heavy
	Very Short	H	SH	LH	U	U
	Short	SH	H	SH	LH	U
	Medium			H	LH	U
	Tall			H	SH	U
	Very Tall	U	LH	H	SH	LH

Example #1 (7/13)

Step 4: Activate Rules

$$\mu_{\text{height}} = \{\mu_{VS} \quad \mu_S \quad \mu_M \quad \mu_T \quad \mu_{VT}\}$$

$$\mu_{\text{height}} = \{0 \quad 0 \quad 0.7 \quad 0.3 \quad 0\}$$

$$\mu_{\text{weight}} = \{\mu_{VS} \quad \mu_S \quad \mu_M \quad \mu_H \quad \mu_{VH}\}$$

$$\mu_{\text{weight}} = \{0.8 \quad 0.2 \quad 0 \quad 0 \quad 0\}$$

Fuzzy Rules Table

		Weight				
Height		Very Slim	Slim	Medium	Heavy	Very Heavy
	Very Slim	H	SH	LH	U	U
	Slim	SH	H	SH	LH	U
	Medium			H	LH	U
	Tall			H	SH	U
	Very Tall	U	LH	H	SH	LH



Substitute
membership values

		Weight				
Height		0.8	0.2	Medium	Heavy	Very Heavy
	Very Slim	H	SH	LH	U	U
	Slim	SH	H	SH	LH	U
	0.7	LH	H	H	LH	U
	0.3	U	SH	H	SH	U
	Very Tall	U	LH	H	SH	LH

Example #1 (8/13)

Step 4: Activate Rules

$$\mu_{\text{height}} = \{\mu_{VS} \quad \mu_S \quad \mu_M \quad \mu_T \quad \mu_{VT}\}$$

$$\mu_{\text{height}} = \{0 \quad 0 \quad 0.7 \quad 0.3 \quad 0\}$$

$$\mu_{\text{weight}} = \{\mu_{VS} \quad \mu_S \quad \mu_M \quad \mu_H \quad \mu_{VH}\}$$

$$\mu_{\text{weight}} = \{0.8 \quad 0.2 \quad 0 \quad 0 \quad 0\}$$

Fuzzy Rules Table

		Weight				
Height		0.8	0.2	Medium	Heavy	Very Heavy
	Very Short	H	SH	LH	U	U
	Short	SH	H	SH	LH	U
	0.7	LH	H	H	LH	U
	0.3	U	SH	H	SH	U
	Very Tall	U	LH	H	SH	LH



Perform
min operation

		Weight				
Height		0.8	0.2	Medium (0)	Heavy (0)	Very Heavy (0)
	V. Short (0)	0	0	0	0	0
	Short (0)	0	0	0	0	0
	0.7	0.7	0.2	0	0	0
	0.3	0.3	0.2	0	0	0
	Very Tall (0)	0	0	0	0	0

Example #1 (9/13)

Step 5: Compute decision function

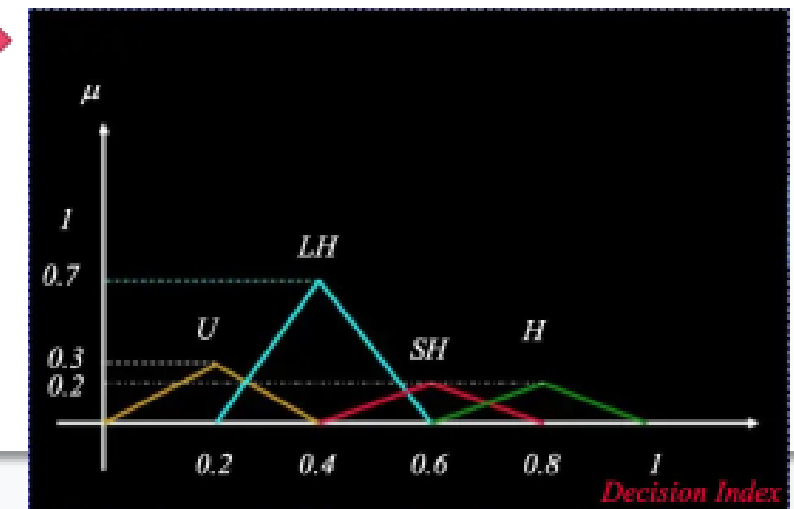
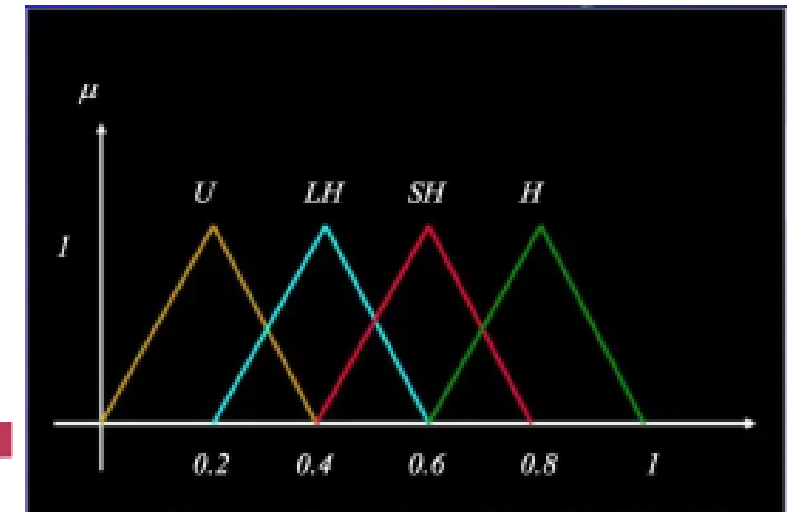
Weight			
Height		0.8	0.2
	V. Short (0)	0	0
	Short (0)	0	0
	0.7	0.7	0.2
	0.3	0.3	0.2
	Very Tall (0)	0	0

Weight			
Height		0.8	0.2
	V. Short (0)	H	SH
	Short (0)	SH	H
	0.7	LH	H
	0.3	U	SH
	Very Tall (0)	U	LH

$$\mu = \{ U \quad LH \quad SH \quad H \}$$

$$\mu = \{ 0.3 \quad 0.7 \quad 0.2 \quad 0.2 \}$$

Scaled
fuzzied
decision



Example #1 (10/13)

Step 6: Compute Final Decision

Use the fuzzified rules to compute the final decision

- Two methods are often used
 - Maximum Method
 - Centroid

Example #1 (11/13)

Step 6: Compute Final Decision using Max Method

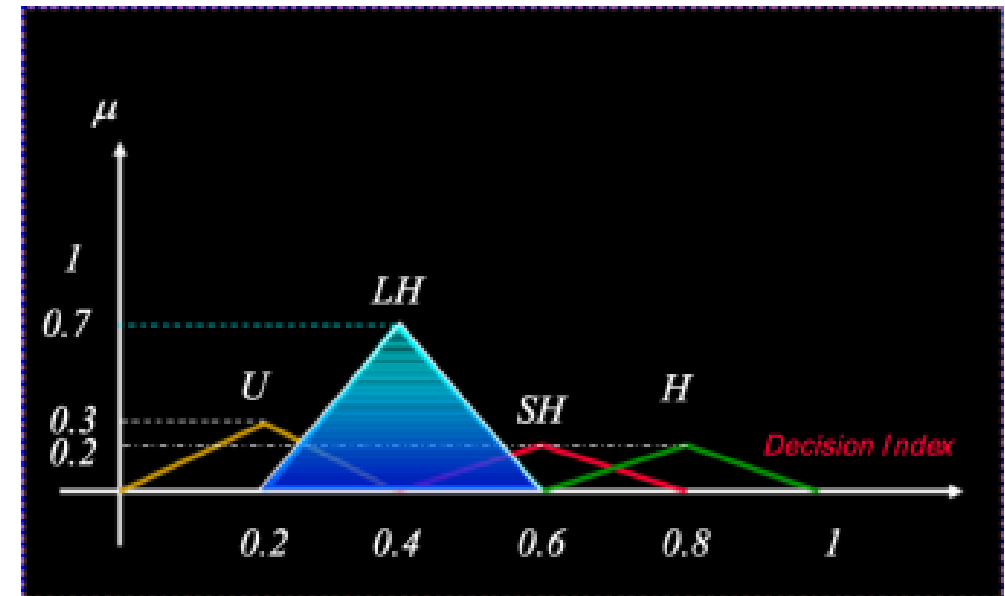
Fuzzy set with the largest membership value is selected

- Fuzzy decision:

$$\mu = \{ U \quad LH \quad SH \quad H \}$$

$$\mu = \{ 0.3 \quad 0.7 \quad 0.2 \quad 0.2 \}$$

- Final decision = Less Healthy
- If two decisions have same membership max, use the average of the two



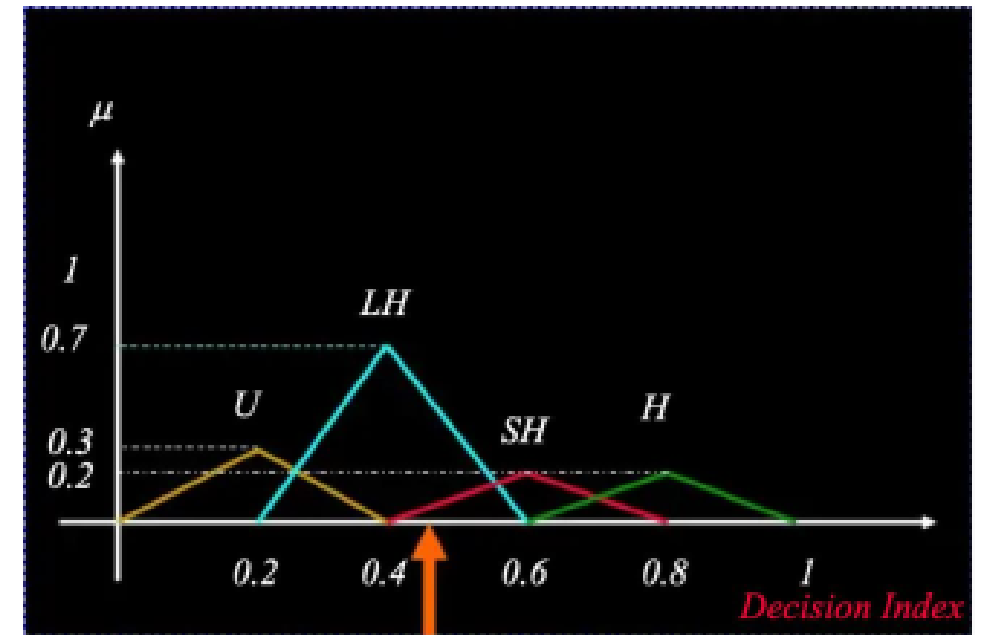
Example #1 (12/13)

Step 6: Compute Final Decision using Centroid Method

$$\begin{aligned} \bullet \quad FD &= \frac{\sum \mu \cdot D}{\sum \mu} = \frac{\mu_U D_U + \mu_{LH} D_{LH} + \dots}{\mu_U + \mu_{LH} + \dots} \\ FD &= \frac{0.3 \times 0.2 + 0.7 \times 0.4 + 0.2 \times 0.6 + 0.2 \times 0.8}{0.3 + 0.7 + 0.2 + 0.2} \\ &= 0.4429 \end{aligned}$$

- Crisp Decision Index (D) = 0.4429

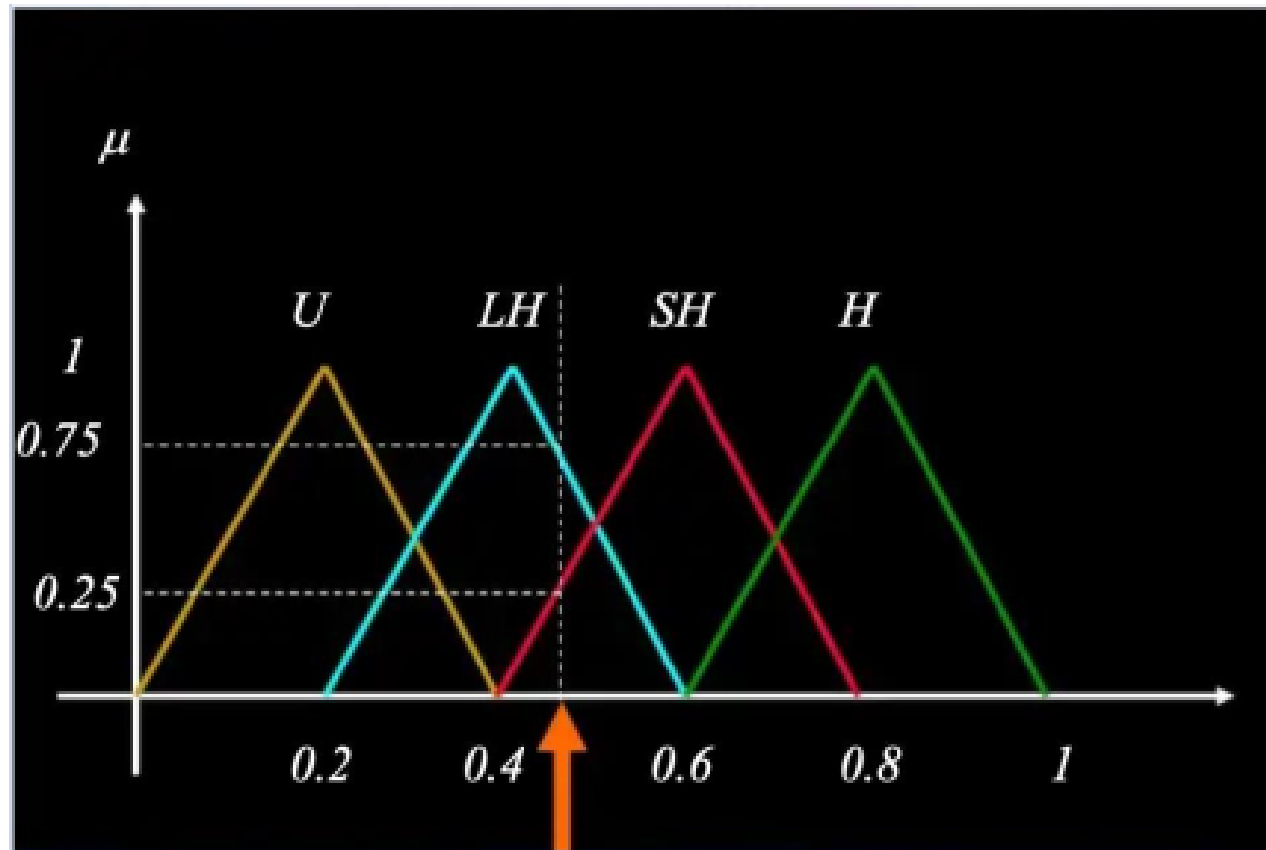
Crisp decision Index (D) is the Centroid



Fuzzified decision

Example #1 (13/13)

Step 6: Compute Final Decision using Centroid Method



Fuzzy decision Index (D)

75% in Less Healthy group

25% in Somewhat Healthy group

Example #2

Example #2 (1/8)

Assume that we need to evaluate student applicants based on their GPA and GRE scores

- For simplicity, let us have three categories for each score

High (H) , Medium (M) , Low (L)

- Let us assume that the decision should be

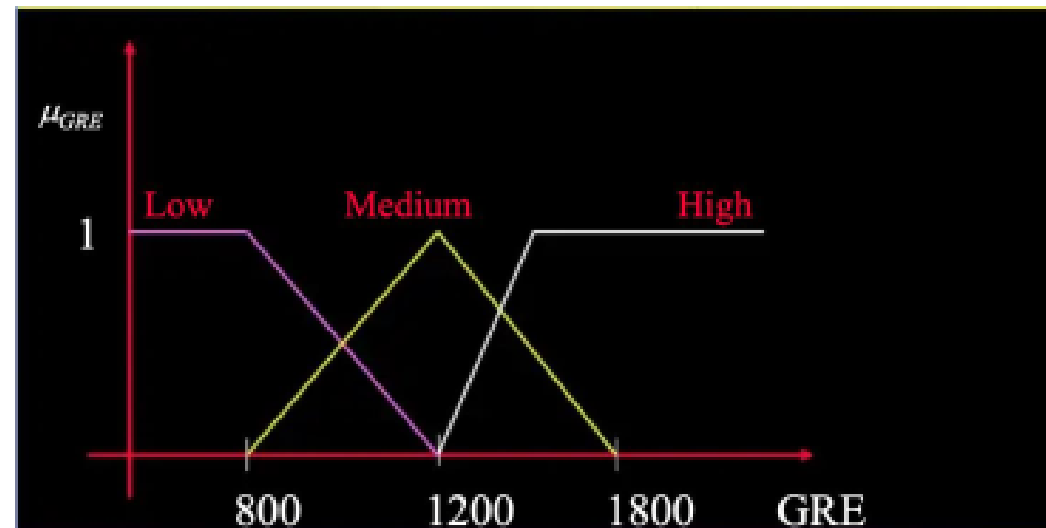
Excellent (E) , Very Good (VG) , Good (G) , Fair (F) , Poor (P)

- An expert will associate the decisions to the GPA and GRE score. They are then Tabulated

Example #2 (2/8)

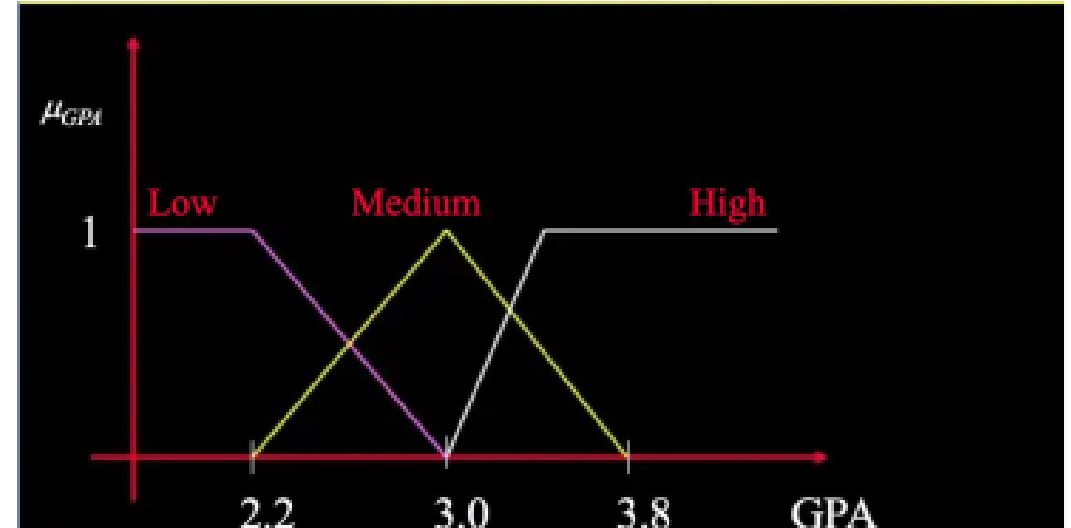
Membership functions

Membership function of GRE



$$\mu_{GRE} = \{\mu_L, \mu_M, \mu_H\}$$

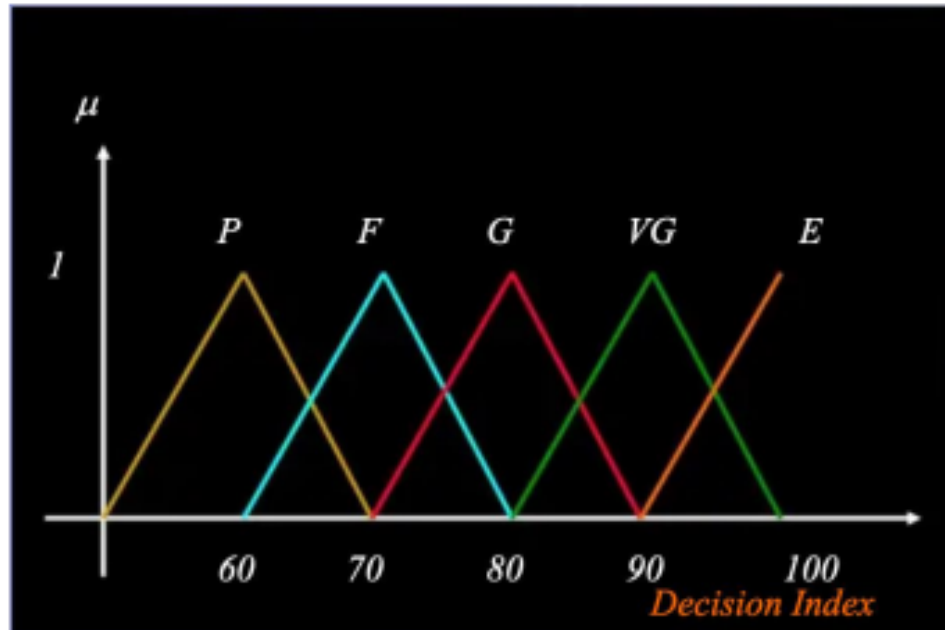
Membership function of GPA



$$\mu_{GPA} = \{\mu_L, \mu_M, \mu_H\}$$

Example #2 (3/8)

Fuzzified Decision



$$\mu_D = \{\mu_P, \mu_F, \mu_G, \mu_{VG}, \mu_E\}$$

Fuzzy Rules Table

GRE				
GPA		H	M	L
	H	E	VG	F
	M	G	G	P
	L	F	P	P

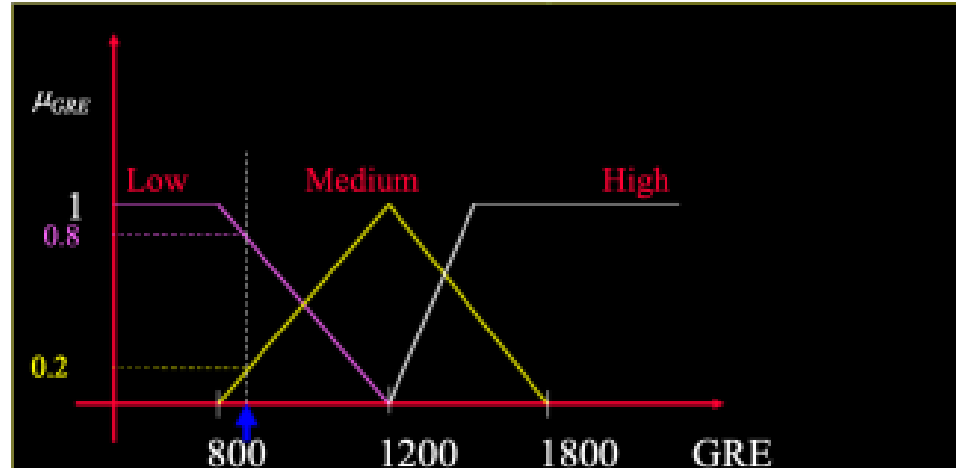
Example #2 (4/8)

Calculate

A decision on the classification of the applicant is needed

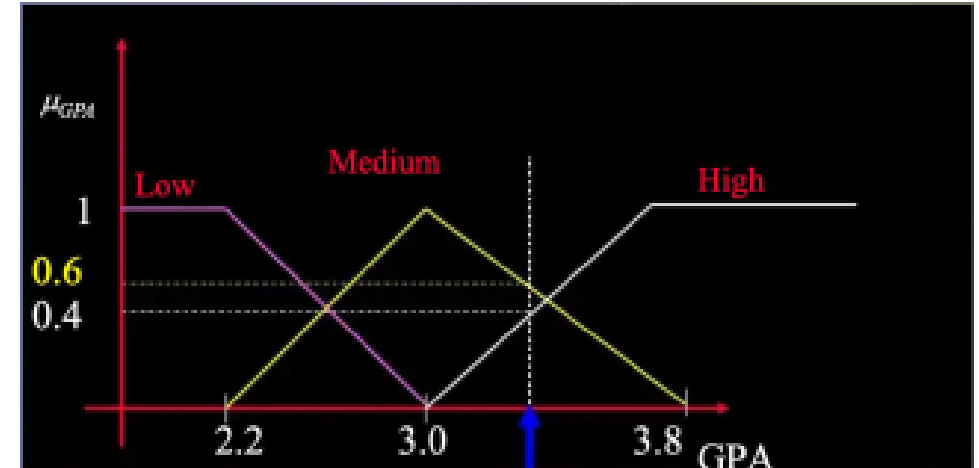
- Assume a student with GRE = 900
- Assume a student with GPA = 3.6

Membership of GRE



$$\mu_{GRE} = \{\mu_L = 0.8, \mu_M = 0.2, \mu_H = 0\}$$

Membership of GPA



$$\mu_{GPA} = \{\mu_L = 0, \mu_M = 0.6, \mu_H = 0.4\}$$

Example #2 (5/8)

Activate Rules

$$\mu_{GRE} = \{\mu_L = 0.8, \mu_M = 0.2, \mu_H = 0\}$$

$$\mu_{GPA} = \{\mu_L = 0, \mu_M = 0.6, \mu_H = 0.4\}$$

GRE				
GPA		H	M	L
	H	E	VG	F
	M	G	G	P
	L	F	P	P

GRE				
GPA		H	M	L
	H	E		
	M	G		
	L	F	P	P



Example #2 (5/8)

Activate Rules

$$\mu_{GRE} = \{\mu_L = 0.8, \mu_M = 0.2, \mu_H = 0\}$$

$$\mu_{GPA} = \{\mu_L = 0, \mu_M = 0.6, \mu_H = 0.4\}$$

GRE				
GPA		0	0.2	0.8
	0.4	0	0.2	0.4
	0.6	0	0.2	0.6
	0	0	0	0

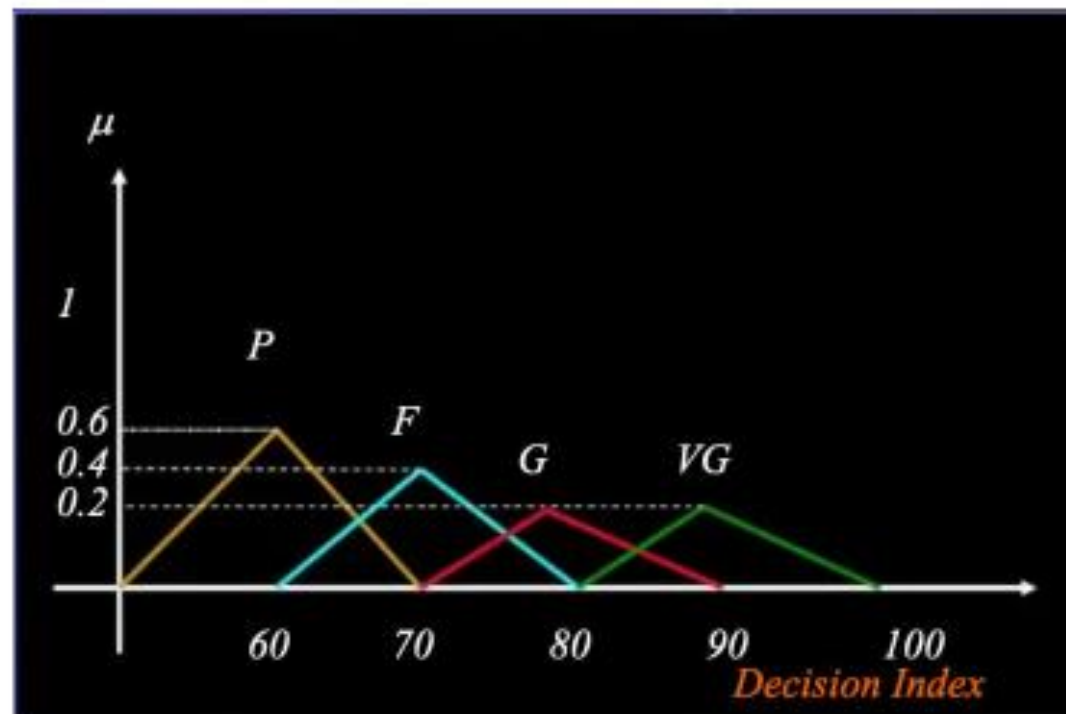
GRE				
GPA		H	M	L
	H	E	VG	F
	M	G	G	P
	L	F	P	P

$$\mu_D = \{\mu_P, \mu_F, \mu_G, \mu_{VG}, \mu_E\}$$

$$\mu_D = \{\mu_P = 0.6, \mu_F = 0.4, \mu_G = 0.2, \mu_{VG} = 0.2, \mu_E = 0\}$$

Example #2 (6/8)

Scaled fuzzified decision



$$\mu_D = \{\mu_P = 0.6, \mu_F = 0.4, \mu_G = 0.2, \mu_{VG} = 0.2, \mu_E = 0\}$$

Example #2 (7/8)

Defuzzification using Max Method

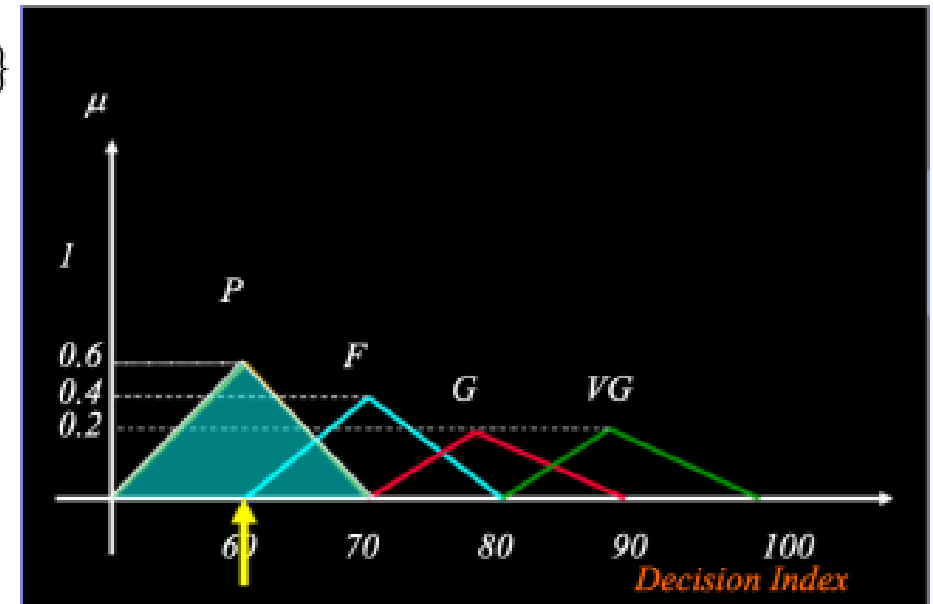
Fuzzy set with the largest membership value is selected

- Fuzzy decision

$$\mu_D = \{\mu_P, \mu_F, \mu_G, \mu_{VG}, \mu_E\}$$

$$\mu_D = \{\mu_P = 0.6, \mu_F = 0.4, \mu_G = 0.2, \mu_{VG} = 0.2, \mu_E = 0\}$$

- Final decision = Poor Student



Example #2 (8/8)

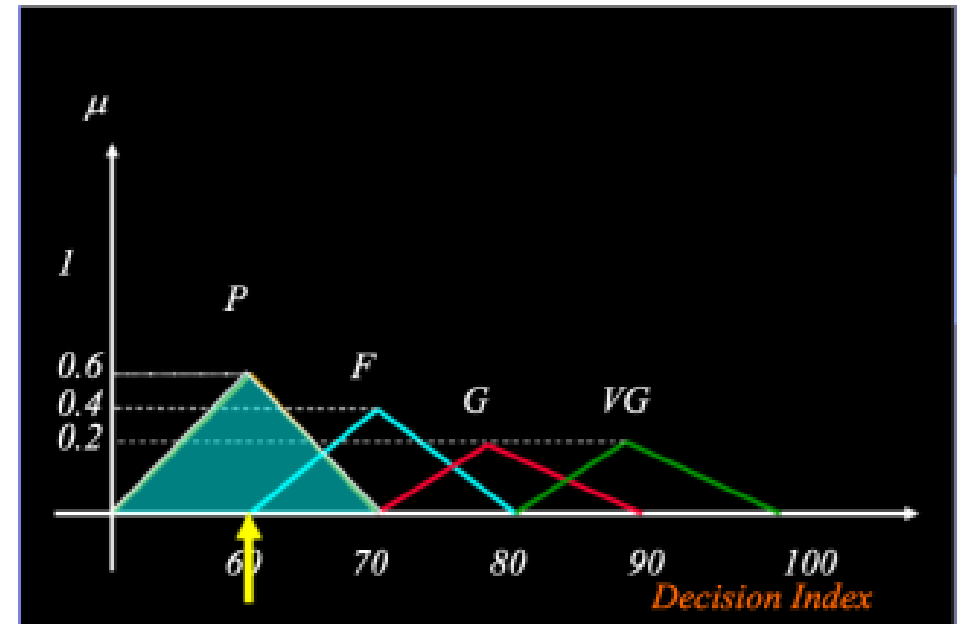
Defuzzification using Centroid Method

$$\bullet FD = \frac{\sum \mu \cdot D}{\sum \mu} = \frac{\mu_E D_E + \mu_{VG} D_{VG} + \dots}{\mu_E + \mu_{VG} + \dots}$$

$$FD = \frac{0.0 \times 100 + 0.2 \times 90 + 0.2 \times 80 + 0.4 \times 70 + 0.6 \times 60}{0.2 + 0.2 + 0.4 + 0.6}$$

$$= 70$$

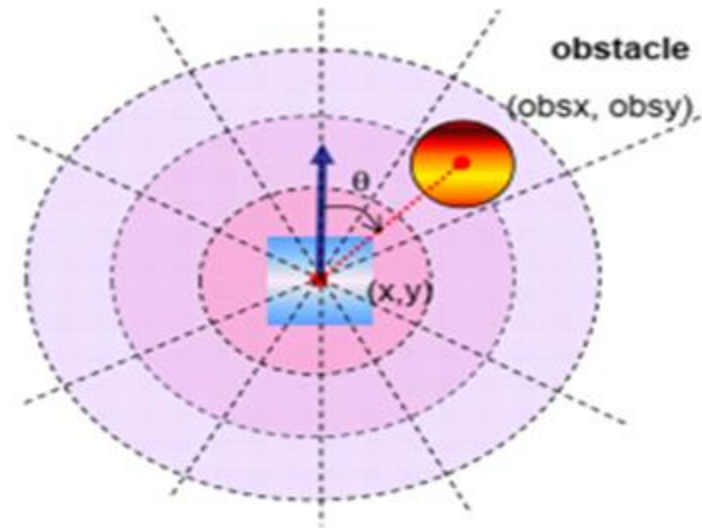
- Final Decision (FD) = Fair student



Practice

Finding Target

Robot Navigation



Obstacle Avoidance & Target Pursuit



Demonstration

Can you describe how the robot should turn based on the position and angle of the target ?



Rule Systems (Steering)

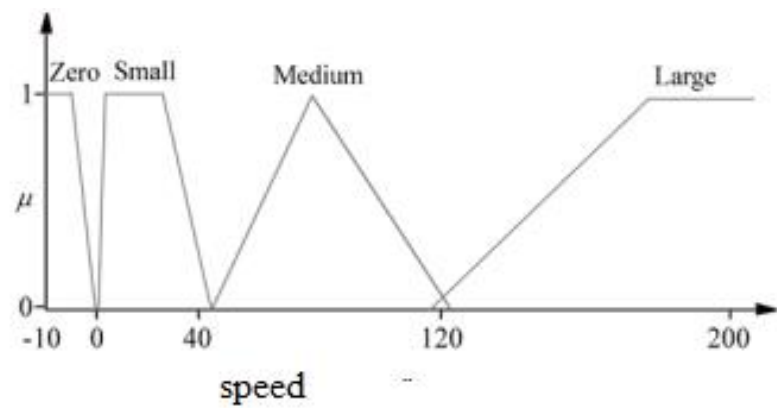
D I S T A N C E	Angle			
		Negatif	Zero	Positif
	Near	Left	Zero	Right
	Medium	Left	Zero	Right
	Far	Left	Zero	Right

Rule Systems (Speed)

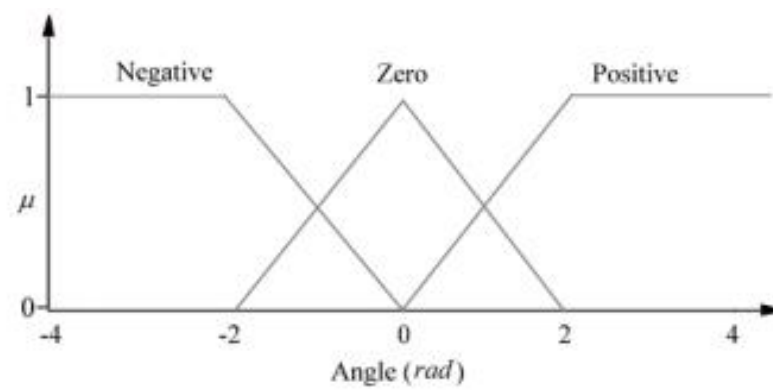
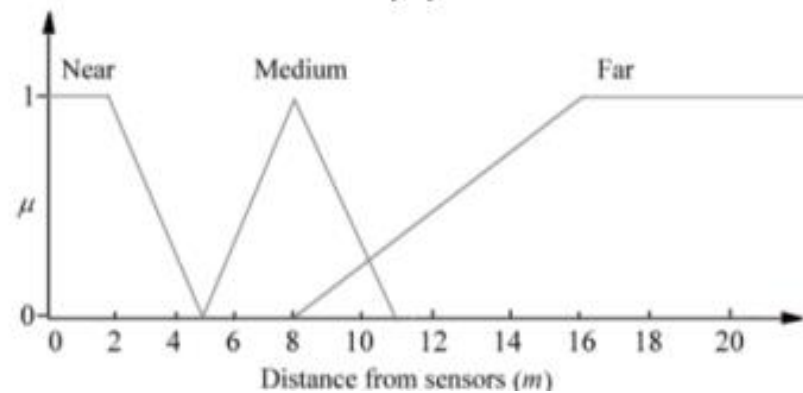
D I S T A N C E	Angle			
		Negatif	Zero	Positif
	Near	Small	Zero	Small
	Medium	Medium	Medium	Medium
	Far	Large	Large	Large

Question

1. How much is the speed and the steering-angle, if the distance between robot and obstacle is 15 m, and the angle from the obstacle is 45 degree



(a)



(b)

